Life Cycle Analysis of 3 Utilisation Pathways in the Production of Biogas/Bioenergy by the Anaerobic Digestion of Food Waste in Northern Ireland

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Individual Contributions

Table 1: Statement of Individual Contributions

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N.B. (1) Contributions in parenthesis denote work under duplicate titles submitted separately to this study that is not featured in this paper.

N.B. (2) Shaded line-items denote previous contributing members who contributed intangibly to the undertaking of this study, but without notable significance to the works herein.

Definitions

AD	Anaerobic Digestion
AFBI	Agri-Food and Biosciences Institute
CEAP	Circular Economy Action Plan
CHP	Combined Heat and Power
COP26	26th Conference of the Parties (26th UN Climate Conference)
CSTR	Continuous Stirred Tank Reactor
CTP	Climate Target Plan
EC	European Commission
EP	European Parliament
EU	European Union
FIT	Feed-in Tariff
FU	Functional Unit
FW	Food Waste
GHG	Greenhouse Gas
G-t-g	Gas-to-grid
GVA	Gross Value Added
HGV	Heavy Goods Vehicle
LCA	Life Cycle Analysis
LNG	Liquified Natural Gas
NDC	Nationally Determined Contributions
NI	Northern Ireland
NIRO	Northern Ireland Renewables Obligation
NTS	National Transmission System
OMSW	Organic Fraction of Municipal Wastes
PFD	Process Flow Diagram
REDII	Renewable Energy Directive (EC)
RHI	Renewable Heat Incentive
RNG	Renewable Natural Gas
ROC	Renewables Obligation Certificates
RTFO	Renewable Transport Fuel Obligation
SHW	Slaughterhouse Waste
SMWH	Slaughterhouse Waste Mixed at Annual Production Levels
SO	Soft Offal
SW	Slaughterhouse Waste
TRL	Technology Readiness Level
UK	United Kingdom
UN	United Nations
VFA	Volatile Fatty Acids
WI	Wobbe Index

1. Introduction

The escalating demand for the reduction of global greenhouse gas (GHG) emissions is reflected in its primacy as a feature of geopolitical discourse [1, p. 2-4, 2, 3]. In a historic effort to consolidate this discourse into proportionate strategic advancement, the United Kingdom (UK) will host the 26th United Nations (UN) Climate Conference (COP26) in November 2021 [4, p. 11-15]. Member states and industry leaders are predicted to intensify their commitments to global temperature targets set in the UN's 2016 Paris Agreement [5] by pursuing enhanced 'economy-wide absolute emission reduction targets' [5, p. 4] for the first time since the agreement was signed [6]. Sincerely-held pledges to this course of action will have a significant impact on existing derivative initiatives such as the European Union's (EU) 2019 European Green Deal [7, Section 3], and consequently, member-state domestic and industrial policy (constituted by the recently enshrined 2020 European Climate Law [8]).

Practicably, [7] builds upon previous emissions reductions strategies that have proven the European Commission's (EC) commitment to the Paris climate target. In 2007, it set out the 20:20:20 target [9]:

- reduce GHG emissions by 20% (relative to 1990s measures)
- incorporate 20% renewable produced energy
- achieve 20% increase in efficiency

- the target was achieved on all counts [10] **(see appendix A)**. Subsequently, the European Parliament (EP) published both an intermediary Climate Target Plan (CTP) for 2030 [11] and a stretch target of net-zero GHG emissions (climate neutrality) by 2050 in [7], to which member states must align their UN mandated nationally determined contributions (NDCs) [12].

Globally, energy production and agriculture account for the largest contributions to GHG emissions by economic sector [13] **(see appendix B)**. Respectively, the EC published a number of revisions to its existing secondary directives to accommodate the enhanced targets set out in [11] and [7], the most pertinent of which is the 2021 amendment [14] to the Renewable Energy Directive (REDII) [15]. [15, L 140/17] dictates a boost in biogas installation investment (citing a large potential for reductions in GHG emissions as well as

sustainability and income benefits to farmers and rural areas), which serves as a precursor to the UK's 2011 Anaerobic Digestion (AD) Strategy and Action Plan [16], and its 2012 Bioenergy Strategy [17] (due for revision in 2022 [18]). These strategies capitalise on the development of the biorefinery concept (analogous to petrol refineries) which the IEA describes as 'the sustainable processing of biomass into a spectrum of marketable products and energy' [19]. Despite the UK's departure from the EU in 2020 [20], its pre-existing legislative infrastructure and continued alignment with the UN's targets influence much of its emerging net-zero workstreams [21], exemplified by its direct amendment of [15] to keep REDII enshrined in UK law [22].

Northern Ireland (NI)

Under current projections, NI will struggle to meet the UK's GHG reduction targets with projected figures of ~35% in 2019 compared to the 57% CTP target for 2030 [23]. In the same year, 26% of NI's GHG emissions were attributed to the agricultural sector [24] compared to 10% for the UK [25], which mirrors its disproportionate contribution to the NI economy (the agri-food sector in NI represents 1.7% of NI's gross value added (GVA) compared to 0.5% for the UK [26, p. 6]). Finally, despite its own recommendation to increase investment into agricultural waste-to-energy initiatives for the reduction of the sector's impact on the environment, there is a growing recognition that the implementation of the NI Agri-Food Strategy Board's 2013 'Going for Growth' strategy [27, p. 15] has burdened the region with significant environmental costs in its aims to increase the sector's market and employment share [27, p. 4]. In 2018, NI produced 10.3 tonnes of CO_2 per person compared to 6.8 tonnes per person for the UK [28].

In compatibility with UK policy, the Northern Irish Executive published its derivative Strategic Energy Framework in 2010 (due for revision in 2021 [29]) and its GHG Emissions Reduction Action plan the following year [30]. These frameworks emphasise the opportunities for AD utilisation within the developing biorefinery concept in the reduction of GHG emissions (through the production of biogas/bioenergy from biofuels), and underscore the development of new and existing incentives to increase the uptake of technologies such as AD in the region [31]. Such incentives and political instruments relevant to this study include (but are not limited to):

- *The Northern Ireland Renewables Obligation (NIRO*) – an obligation for electricity suppliers to generate a minimum proportion of renewable energy [32]

- *Renewables Obligation Certificates (ROCs)* – certificates produced by suppliers representing the proportion of electricity generated through renewables via different technologies [32] **(see appendix C)**
- *Renewable Transport Fuel Obligation (RTFO)* - an obligation imposed by the UK Government onto fuel suppliers to supply a minimum proportion of renewable or sustainably sourced fuel [33]

N.B. The Renewable Heat Incentive (RHI) [34] and Feed in Tariff (FIT) [35] have been excluded from commentary in this section as they are no longer applicable in this region, but will be referred to subsequently in this study as when relevant.

Circular and Bioeconomy in NI

As well as the need to generate renewable energy for the reduction of GHG emissions, another key driver for the exploration of bioenergy, the biorefinery concept and AD technologies is the environmental burden of waste [36, p. 3]. Recognised as fundamental to achieving the European Green Deal's climate neutrality target [37], the EC's 2020 Circular Economy Action Plan (CEAP) aims to address the loss of 90% of biodiversity and water stress as well as 50% of total GHG emissions attributed to resource extraction and processing [38]. Accordingly, the UK has accepted the EU's amendments to 2 relevant directives which inform regional waste management law: the Waste Framework Directive and the Landfill of Waste directive, both of which feature requirements for enhanced recovery and re-use of waste, and diversion of waste from landfill sites [39] (requirements facilitated in the development of the biorefinery concept through AD technologies as illustrated in [16] and [17]). Particularly, these policy drivers impact the management of food waste (FW) due to the development of recent secondary domestic legislation banning the landfilling of source separated FW and mandating the separation of household waste [36].

NI's anticipated 'Environmental Strategy for Northern Ireland' is expected to address the enhancements required by [38], building upon the Executive's 2013 'Delivering Resource Efficiency' strategy which advises both the adoption of AD technologies for the improved recovery of waste in NI (in aiming towards a circular economy), and a life cycle approach to waste management [40] as is adopted in this study. While policy initiatives focused on eco-efficiency and waste management facilitate the circular economy agenda, research into a regional bioeconomy focuses primarily on the sustainable exploitation of biomaterials and is referred to as a subsection of pre-existing circular economy policy briefing due to its

overlapping drivers [41, p. 11-12]. Resulting research determined that AD technology is a valuable model through which to inform the development of bioeconomy policy in NI [42]. Finally, the region's recently published agricultural Innovation Strategy presents a clear pledge to invest in a 'Transformative Bioeconomy' to mitigate the GHG emissions and burden of waste attributable to the agri-food sector through R&D initiatives in such relevant industries [43], establishing a clear context for this study.

Background (Advancements in AD)

AD is one of the most common biochemical processes in the developing biorefinery concept [44], describing a series of processes converting biomaterials into biogas and digestate by micro-organisms in the absence of oxygen [45]. Literature typically identifies the following 4 AD process steps: hydrolysis, acidogenesis, acetogenesis and methanogenesis [46].

Table 2: Examples of raw biogas compositions

	Source[a]			
	Kapoor et al (2020)[47]	Benato et al (2019)[48]	Kuo et al (2017)[49]	Muñoz et al (2015)[50]
	Primary Feedstock[b]			
	(Average)	Cattle sewage	Commercial Food Waste	Landfill
Constituent	**Concentration (v/v)**			
Methane	40-60%	53 ±8%	64.6%	35-65%
Carbon Dioxide	35-55%	44 ±7%	34.9%	15-50%
Moisture	1-5%	<2%	-	<5%
Nitrogen	0-5%	1.9±0.3%	0.7%	5-40%
Hydrogen	Trace	<0.5%	<1.3%	<3%
Hydrogen Sulphide	0.1-3%	645ppm	163ppm	<100ppm
Oxygen	<2%	0.4±0.06%	<0.1%	<5%
Trace gases	<2%	<2%	<2%	<250ppm
Ammonia	0-500ppm	15±4.7ppm	Trace	<5ppm

[a] Sources sampled from both industrial examples and simulation-based approaches

[b] Co-digestion employed in all cases where a feedstock is identified, the most significant of which (according to the authors) is displayed above

[c] Calculated by conversion for consistency

The production of biogas is consequential to the aim of reducing GHG emissions; despite being composed largely of GHG (and as such presents itself as a multi-faceted fuel source, for example being capable of replacing fossil fuels such as natural gas in incumbent energy systems), its components are considered biogenic and carbon-neutral [47]. Notably, the

biogenic methane (biomethane) constituent of natural gas is also referred to as renewable natural gas (RNG) , reflecting its production from non-anthropogenic waste material that would otherwise present a net carbon burden if not diverted from incineration or landfill as per circular economy theory [51].

N.B. Although the constituents of biogas are considered carbon-neutral, the added burdens of the production process do incur a cost of equivalent carbon utilisation which serves as the focus of this study.

Expressly, AD technologies represent a waste-management opportunity in the recovery of waste belonging to 3 categories of biomass: farm substrate (manure, energy crops, feed waste), waste from private households or municipalities (organic fraction of municipal wastes (OMSW), food waste, market waste) and industrial by-products [52, p. 1]. FW is of particular interest due to its considerably higher methane yield (on average) as illustrated in Table 1, as well as its intersection with the high GHG impact agricultural sector in the form of slaughterhouse waste (SHW), and its relevance to the recently developed waste-management policy drivers previously referenced.

As a product, the multi-faceted quality of biogas is illustrated below by a (non-exhaustive) representation of its utilisation pathways through a schematic of a large-scale AD operation:

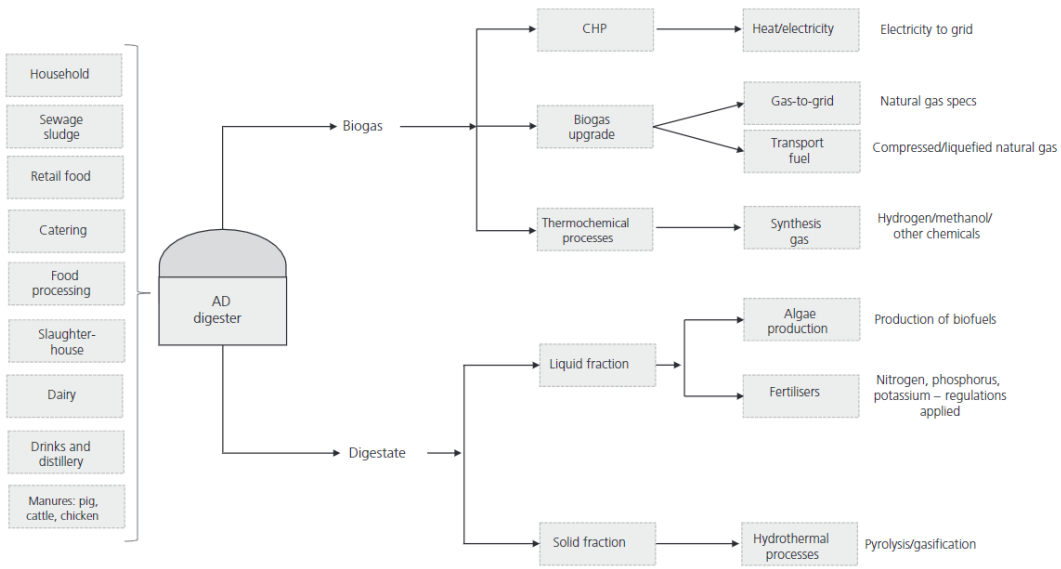

Figure 1: Feedstocks and utilisation pathways for AD products in a large-scale biorefinery [42]

As illustrated, there are 4 main utilisation pathways (scenarios) for raw biogas [36, 52, 53]:

- *scenario 1*: combustion in a combined heat and power plant (CHP) to produce heat and power

- *scenario 2*: upgrading to biomethane for injection into the grid
- *scenario 3*: compression of upgraded biomethane into biodiesel/liquified natural gas (LNG)
- *scenario 4*: processing to chemicals via syngas

The remaining digestate fractions present further value to the development of a bioeconomy in its applications due its high macro and micronutrient density [54].

AD is an established technology with demonstrated impact in GHG emission reduction: Vulcan Renewables Biomethane Plant has converted 70,000 tpa of energy crops and silage since its commissioning in 2013, resulting in 3265 tonnes CO_2 equivalent GHG emission reductions (900m^3/hr biomethane) [55].

Furthermore, it is estimated that 2018 biogas production levels could sustain almost 20% of the world's gas demand, with Europe at the forefront of production [56]. In the same year, there were 660 AD plants in the UK (51% classed as agricultural) producing 3.9% of energy consumed from natural gas [57]. Regionally, AD has a projected worth of £33M to the NI economy [58], and there are currently 76 AD plants present (more per capita and square kilometre than the rest of the UK). Although the initial uptake of AD technology in NI was comparatively slow, the region is considered the most attractive region in the UK for AD due to its high availability of feedstocks and government support schemes. [59, p. 4]

The proof of concept providing the basis and regulatory framework for AD (the Holsworthy Project) was presented to the UK Government in 2005 [60]. Although the subsequent Europe-wide waste management/GHG reduction incentives described herein were substantial drivers for the development of AD technologies (a recent review of technological development throughout Europe shows a substantial increase in AD patents since the 2000s [61], technical development and knowledge transfer regarding AD resource management, digestion conditions and feedstock variability still forms a critical component of ongoing research [42].

Exemplifying the complexity of analysing the value of the AD biorefinery within regional bio-economies, the potential for GHG emission reductions shows great variability under influence by the following (interconnected) factors [36]:

- land-use management practices
- feedstock/s
- biogas yield

- plant management
- efficiency (e.g., methane slip)
- product utilisation

Earlier studies

Previous research has explored a range of scenarios with the aim of quantifying the GHG emission reduction potential of AD technologies on the island of Ireland. [36] explored the substitution of food waste for traditional AD feedstocks and its effect on GHG emission reductions within the NI biorefinery concept, while [62] explored product utilisation pathways through the lens of exhaustive feedstock quantification. [63] provides an analysis of methane yields from a range of constituent cattle slaughterhouse wastes in the calculation of energy recovery from the CHP utilisation pathway (scenario 3 as described herein), while [42] identified non-CHP utilisation pathways for biogas and provided a quantification of and applications for digestate within the NI bioeconomy. Most pertinently, [64] analysed the GHG emission reduction potentials of the 3 scenarios under evaluation in this study using traditional feedstocks based on an operational plant within the NI region. Currently, the predominant feedstock for AD plants in NI is animal waste, and the least used feedstocks are dairy waste, effluent, and animal processing waste, providing opportunity for evaluation in this study.

Feedstocks used by AD in NI

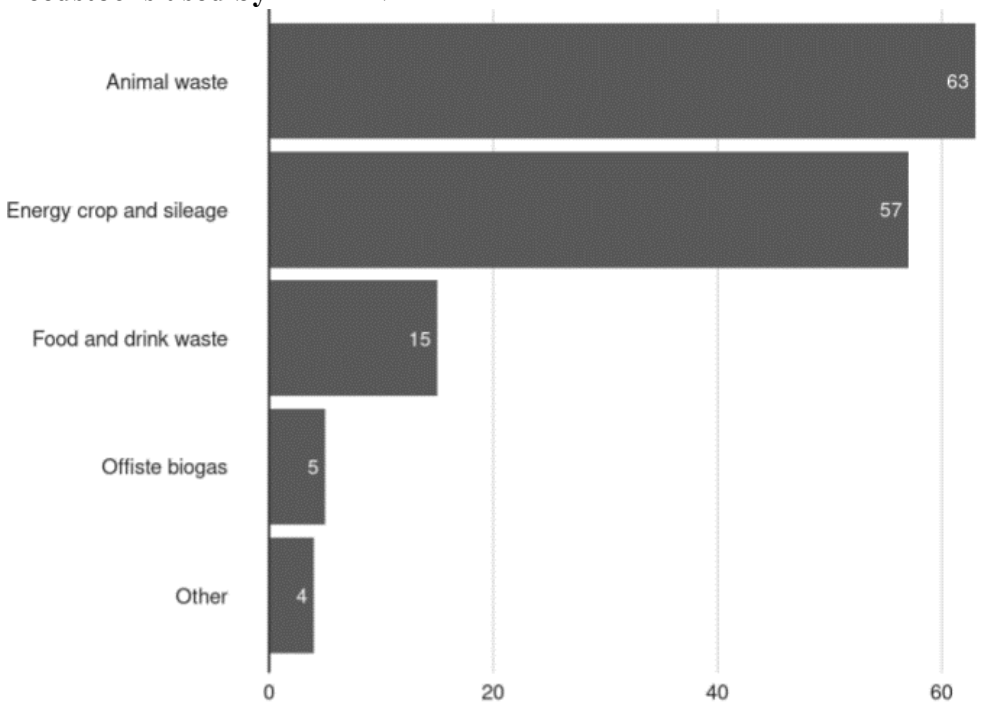

Figure 2: AD feedstock usage in NI.

N.B. 1 Animal waste - Animal slurries and litter (includes cattle, pig, poultry and unspecified), Energy crop and silage (includes energy crops, maize and grass silage), Food and drink waste (includes municipal, commercial and industrial), Offsite biogas (includes containerised offsite biogas), Other (includes dairy waste, effluent and animal processing waste). Note that 1 AD facility's feedstock was unknown and has such not been included. Data extracted from NNFCC biogas map [65].

N.B. 2 Although animal processing waste is included within the 'other' category of the above graph, this study considers it FW.

Although previous research has explored both the substitution of FW as a feedstock, and the effect of utilisation pathways on energy recovery, [36] does not further explore variability within FW sources and their effects on GHG emission reductions, or the impact of varying utilisation pathways (results are modelled on an operational CHP plant). [62] provides a broad, theoretical overview of FW feedstock comparisons layered with a biogas utilisation pathway analysis, but the two analyses do not intersect, and the resultant recommendations are not directly practicable. While [63] refers to a specific FW feedstock for analysis, it also models its results on a CHP plant and does not consider the impact of product utilisation on energy recovery. Finally, although [64] does reflect an operational plant within the scope of the 3 scenarios under evaluation in this study, its usage of traditional feedstocks in a currently operational plant does not allow for the optimisation of operational recommendations for new entrants as regards to unit configuration, operating conditions or feedstock selection.

Aim of This Study

The aim of this study was the evaluation of the cradle-to-grave environmental impacts of 3 biogas utilisation pathways in AD in NI, as well as those impacts of substituting traditional AD feedstocks in an operational plant with SHW (as further justified in section 3.2.1). This was achieved through the pursuit of the following objectives:

- to carry out an analysis of regional SHW feedstocks and their suitability for AD (in the context of an operational biorefinery within the regional bioeconomy)
- quantify potential GHG emission reductions under different unit configurations through scenario modelling for operational recommendations to industry
- compare outputs from the model to current-state AD infrastructure in NI to gain an understanding of the technical, geographical, and political barriers and enablers for different utilisation opportunities for biogas, informing regional bioeconomy policy

Additionally, a sensitivity analysis was undertaken to explore the impact of the following variables on GHG emission reduction potential:

- feedstock selection
- biogas upgrading method

Previous works provide valuable knowledge transfer within the NI bioeconomy through either broad, theoretical simulations or specific, isolated studies, but this study will provide the following novel elements:

- the provision of a practicable assessment for developing regional biorefinery initiatives (which include complete recommendations for unit/process selection, plant location and feedstock procurement relating to transportation distances)
- a targeted analysis of environmental technology facilitating GHG emissions reductions at the intersection between the agricultural sector and food waste production

To further knowledge transfer in the area of on-farm digestion and its life-cycle benefits, the NI Agri-Food & Biosciences Institute (AFBI) established an AD research plant in Hillsborough in 2008 [66], providing an accurate representation of an operational AD plant with a large volume of peer-reviewed experimental data and primary data to serve as a both an input source and comparison to the model in this study [67].

2. Goal and Scope

The goal of this study is the measurement of the environmental impacts of 3 potential utilisation pathways (scenarios) of biogas produced by the AD in NI, as well as to evaluate the environmental impact of substituting traditional feedstocks in an operational plant with agricultural FW (SHW). Scenarios 1-3 (previously illustrated) have been selected for comparison (scenario 4 is not considered within scope as it is the only scenario necessitating technology at a technology readiness level (TRL) below 9 (commercialisation) [62]). The 3 scenarios outlined were compared with 3 baseline scenarios: models based on a modification of current operations of the Hillsborough AFBI plant (which utilises cattle slurry as in [67]). Although the AFBI plant capitalises on 2 AD units (in order to separate the hydrolysis stage and optimise the pH, temperature and retention time for each phase [68, p. 196] and utilises its biogas product through CHP [67, p. 4], the baseline scenario assumes the inclusion of only one AD unit for simplicity and is adapted for all 3 scenarios under evaluation with in-situ post-processing. Furthermore, the operational AFBI plant operates on a series of pre-treatment units to accommodate co-digestion of cattle slurry with silage, with holding tanks included for the controlled addition of feed into the AD. In the base scenario model, the feed is moved through the process continuously (feeding system selected after AD unit selection in section 3.2.2 according to [68, p. 193]) with no buffering, heat, material leakage or power loss through piping or storage is assumed. Furthermore, the baseline pre-treatment process used for comparison is identical to that utilised for the SHW scenarios.

Finally, each model was designed assuming steady-state conditions, so emissions burdens relating to equipment manufacture, land use, maintenance will not be considered within the scope of analysis, as well as the burden of reaching steady state conditions [69].

Methodology

The environmental analysis was conducted through a life cycle analysis (LCA) in accordance with ISO 14040/14044 standard (ISO, 2006) but limited only to the characterisation factor of climate change (energy savings in CO_2 equivalent emissions when compared to natural gras from the grid) due to the time limitation on the undertaking of this study. The economic performance of each process configuration is not included within the scope: in-keeping with the defined aims of this study, a desired outcome of this aim is the recommendation of

workable policy that may reduce the economic barrier to entry for developing technologies that have the greatest impact on lowering the NI agri-food sector's GHG emissions.

In the absence of bespoke LCA software, the scenarios were modelled on excel.

2.1. System Boundaries and Functional Units (FU)

System Boundaries

The overall system boundary begins with the identification and transportation of slaughterhouse waste to the biorefinery and is followed by 4 sub-systems: the AD plant producing raw biogas (including pre-treatment units), and 3 further processes delineating the 3 scenarios under evaluation (processes for treating the raw biogas product for each utilisation pathway).

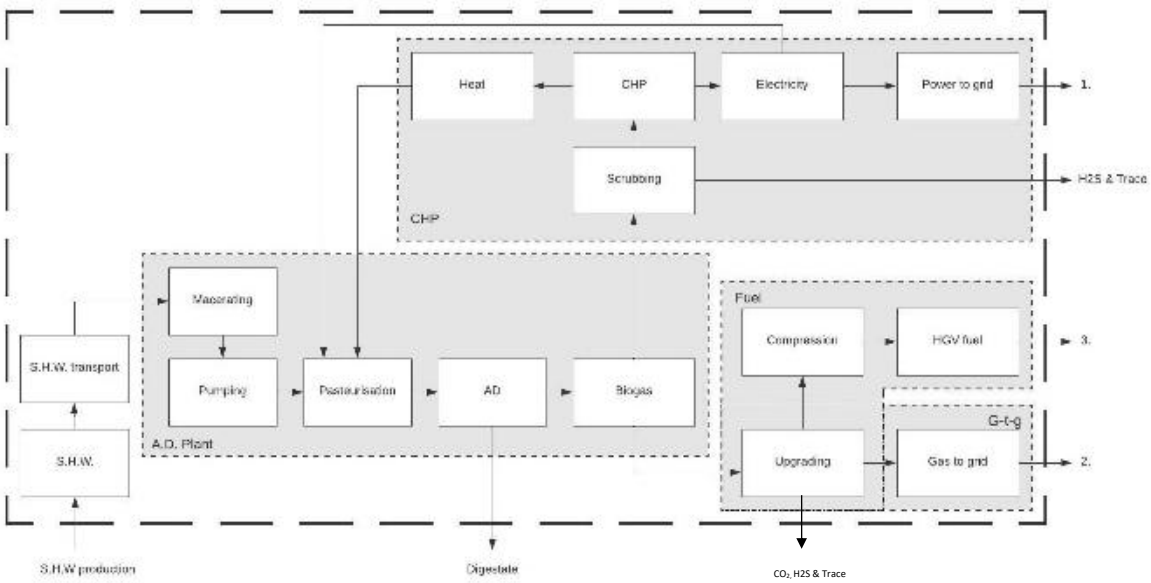

Figure 3: Overall system boundary including all 3 scenarios under investigation. Shaded sections denote an internal boundary for each sub-system under evaluation: the AD plant and systems supporting each of the 3 scenarios ((CHP (1), G-t-g (2), and HGV fuel (3)).

The inclusion of feedstock transportation within the scope builds upon work carried out in [36] which quantifies the impact of feedstock transportation from collection points in NI, asserting that it is the largest contributor to the electricity requirements of processing FW.

Conversely, the production of the feedstock (lifetime GHG emissions) was not considered in scope for this study, as policy recommendations regarding waste reduction on a sectoral basis are considered outside of the remit of this study (the NI agri-food sector (waste producer) was identified as a major stakeholder of this work (alongside relevant policy makers). The sector's stake in the outcomes of this work are put forth by KMPG's Climate Change Bill Impact Assessment, commissioned by NI's minister for agriculture Edwin Poots [70]. The assessment asserted that meeting NI's GHG emissions reduction targets for 2045 would necessitate an 86% reduction in cattle and sheep numbers. In response, the Ulster Farmer's Union published their position on this analysis, proclaiming that investment into sustainable agri-food processing technologies to aid in the reduction of GHG emissions will be vital in defending the sector's viability, which it deems necessary to the welfare of the NI economy [71]. Due to the identification of the sector as a major stakeholder of this work, analysis of the viability of the initial creation of agri-food waste was excluded from this scope.

The selected feedstock is assumed to be homogenous for the simplicity of the model, although in practise a large proportion of the process intensity is dedicated to the stabilisation feedstock composition and properties [69],

AD Plant

Following transportation of the SHW to the biorefinery, the feed is input into the macerator for pre-treatment. The pre-treatment requirements for the AD of SHW are defined by both the following legislative and technical constraints. SHW is classified as a category 3 material according to Article 10 of the EP Regulation (EC) No 1069/2009 which regulates the health rules regarding animal by-products (ABPs) not intended for human consumption [72]. Following Brexit [73], UK biogas plants still operate under EU guidelines. The enforcement of these guidelines in NI is under the jurisdiction of the Animal By-Products (Enforcement) Regulations (Northern Ireland) [74], mandating that category 3 APBs must have a maximum particle size of 12mm). Additionally, physical pre-treatment is necessary to facilitate the rate determining step of AD (hydrolysis), which is accelerated by smaller particle sizes, increasing the surface area to volume ratio of the feedstock particulates (the coefficient of

substrate utilisation doubled when the average particle size decreased from 2.14mm to 1.02mm [46, p. 388]). The particle size constraint together with the requirement of the homogeneity of feedstock for efficient AD [68, p. 193] necessitate the inclusion of a macerating unit (further justified in section 3.2.2).

Furthermore, [72] mandates that category 3 material shall be treated via pasteurisation for 60 minutes at 70°C [72], necessitating the inclusion of the pasteurisation unit as illustrated. It is assumed that 100% of the macerator's output feeds into the pasteuriser. Heat and power (parasitic load from the CHP unit) are also input to the pasteurisation unit.

N.B. In the operation AFBI plant, the heat and power produced in the CHP unit entirely meet the power requirements of the plant rendering it self-sufficient, and there is left-over heat and power to support surrounding buildings and in-situ research projects. However, although the PFD for the models illustrates the parasitic load of the CHP being in-part diverted to the pasteuriser, the model calculations will assume the requirements of the pasteuriser will be met by electricity from the grid.

Following pre-treatment, the SWH is input into the AD unit for digestion (refer to sections 1 and 3.2.2 of this report). The outputs of the AD process are raw biogas and digestate. The environmental impact and potential applications of digestate are considered outside of the scope of this study in recognition of the significant contribution of [75] in this area (in the context of the NI bioeconomy) and the time constraints on undertaking this study.

Raw biogas is considered the sole output of the AD plant within the scope of study (waste treatment of effluent streams is not considered as justified accordingly below), and it is considered the sole input into each of the following scenarios under evaluation:

Scenario 1

During the AD process, H_2S is formed through the degradation of sulphur-containing compounds within the feed [76]. During combustion within the CHP unit, H_2S is oxidised to SO_2: a greenhouse gas and precursor to acid rain which carries a negative environmental impact [77]. Furthermore, the acidity of H_2S will damage combustion equipment in high concentrations [78], necessitating the inclusion of an H_2S scrubbing unit. The outputs of the scrubbing unit are the H_2S removed and the 'clean' biogas. The waste H_2S stream is not considered within the scope of analysis due to its trace quantities within raw biogas composition (refer to table 2) and the complexities of the evaluation of the environmental impacts its treatment as an effluent stream, not feasibly undertaken within the capabilities of this study (presenting an area for further research).

The clean biogas is input into the CHP unit (refer to section 3.2.3), and the outputs are heat and power.

Scenario 2

The substitutability of biogas for natural gas in the National Transmission System (NTS) is dependent on the removal of CO_2 (and other gases) in an upgrading process, resulting in a 98% biomethane product with similar properties to natural gas [52, p. 2] **(see appendix D)**, necessitating the inclusion of an upgrading unit. The output of this unit is biomethane for injection to the grid.

Scenario 3

Biodiesel (LNG) is produced by the liquification of biomethane, rendering an upgrading unit as described for Scenario 2 necessary in the conversion of raw biogas to biomethane. Biodiesel for us as HGV fuel must meet chemical and physical specifications outlined in standard BS EN 14214 [79]. The high pressure required to facilitate the fuel density specified (200 bar) [80] requires the inclusion of a compression unit following the upgrading unit. The sole input into the compressor is upgraded biogas, and the sole output is biodiesel (LNG).

Functional Unit(s)

[36] asserts that precedent in this area supports the inclusion 2 FU (kg and kWh) to facilitate the complexity associated with bioenergy systems (in catering to both waste and energy management stake-holdings). However, the authors argued that their study's focus on GHG emission reductions compared with those of electricity from the grid should be reflected in their selected (singular) FU: 1 MWh of electricity injected to the grid.

Still, as this study aims to provide practicable recommendations to operators of developing biorefinery initiatives, agri-food waste producers and policy makers in its novel approach, a more nuanced consideration of the FU selection was applied.

[81] defines the functional unit as:

FU = [system function] per [unit of calculation] [plus optional: main scenario]

In the context of this study, the function of the biorefinery is to produce power (kWh), and the unit of calculation relevant to both the economic viability of the plant, and the obligations placed upon waste producers through legislation, is feedstock mass (per tonne). The mass of feedstock was selected over the volumetric flowrate to illustrate a workable comparison of feedstock substitution by accounting for density variation, as this relates to the total solids (TS) fraction of the feedstock which in turn influences methane yield [56]. This represents a correct definition of the system, which is highly consequential to reaching the correct conclusions of LCA [81, p. 20].

N.B. Secondary variable units were defined relative to the unit of calculation (gaseous products were calculated in terms of m^3/tonne of feedstock).

FU: kWh/tonne of feedstock.

2.2. Inputs and Outputs

Table 3: Summary table delineating inputs and outputs of each unit illustrated in the system PFD

Process Unit	Inputs	Unit	Outputs	Unit
		AD Plant		
Macerator	SHW	tonne	SHW (macerated)	tonne
Pasteuriser	SHW (macerated)	tonne	SHW (pasteurised)	tonne
	Heat (parasitic load)	kWh/tonne feed		
	Power (parasitic load)	kWh/tonne feed		
AD Unit	SHW (pasteurised)	tonne	Biogas (raw)	m^3/tonne feed
			Digestate	tonne
		Scenario 1		
Scrubbing Unit	Biogas (raw)	m^3/tonne feed	Biogas (clean)	m^3/tonne feed
			H_2S	m^3/tonne feed
CHP	Biogas (clean)	m^3/tonne feed	Heat	kWh/tonne feed
			Power	kWh/tonne feed
		Scenario 2		
Upgrading Unit	Biogas (raw)	m^3/tonne feed	Biomethane	m^3/tonne feed
		Scenario 3		
Compressor	Biomethane	m^3/tonne feed	Biodiesel (LNG)	m^3/tonne feed

3. Literature Review

3.1. Methodology

In each case illustrated below, the literature review adhered to the format of a full systematic review, with the final criteria for selection adhering to a modification of the United Sates Environmental Protection Agency's updated Data Quality Pedigree Matrix for Life Cycle Inventory Data [82] **(see appendix E)**, henceforth referred to as the 'traffic lighting matrix' (**see appendix F)**.

In selecting literature for review, a search in the Queen's University Online Library database and the Google Scholar academic search engine was conducted for academic papers, technical reports and patents indexed since 2017, using relevant keywords as denoted for each case. Next, the data mining process comprised of source loading into Endnote Desktop referencing software and sorting by country and year (constraining for higher scoring in the temporal and geographic correlation criteria in the traffic lighting matrix, providing the basis for limiting the initial search beyond 2017). The standard for geographical correlation was defined as the island of Ireland, and the constraint for operational primary data was defined as the continent of Europe (unless otherwise stated).

During the data mining process, duplicates, non-peer reviewed academic papers and irrelevant sources (as relates to each case) were discarded. The filtering for relevance was achieved by analysis of the frequency of the keywords utilised in the search process via an online, free access tool built for this purpose (http://tagcrowd.com).

From the resultant selection, a maximum of the 6 most relevant (according to the outlined filtering process) were assessed against the traffic lighting matrix criteria and rated accordingly. The highest scoring source for each case was selected for inclusion in the model.

3.2. Scenario Reviews and Data Quality Assessment

3.2.1. Feedstock

Initial search keywords: anaerobic digestion, AD, slaughterhouse waste, poultry waste, cattle waste, food waste, biogas production, industrial food waste, commercial food waste, agri-food waste

Sources assessed against the traffic light matrix: 6 ***(see appendix G)***

Before conducting the initial search for data pertaining specific SWH feedstocks, it was necessary to sequentially establish the following:

- the technical feasibility of using SWH in place of traditional feedstocks
- the extent to which SHW is an environmentally viable feedstock option for AD
- the suitability of the NI region for SWH feeding into AD

The principal components required in AD feedstocks include carbohydrates, proteins, lipids, cellulose, hemicellulose, and lignin. These components are elementally constituted by carbon, nitrogen (comparatively high levels), phosphorus, and sulphur, where the carbon-to-nitrogen (C/N) ratio is essential for the performance of the AD process, but typically low for SWH necessitating co-digestion with other waste categories in practise [83]. Operationally, the constituents of SHW include rumen, stomach and intestinal contents, blood and wastewater representing a high organic content level characterised by a predominance of proteins (12–20% protein/VS) and fats (68–82% fats/VS), indicating high methane yields and therefore its potential suitability as an AD feedstock [84]. Indeed, SHW is consistently reported as having proportionately higher methane yields than other waste categories per unit of mass and high anaerobic biodegradability [75] [84].

Table 4: Characterisation of sampled slaughterhouse waste flow streams [84]

Solid waste and solid mixtures (SW)	Cattle meat and fat	Pig meat and fat	Confiscates (lungs, livers and kidneys)	Pig stomach	Rumen content	SW1	SW2
TS (g kg^{-1})	885.58 ± 4.81	564.54 ± 19.18	244.89 ± 2.89	182.89 ± 7.88	116.86 ± 5.04	527.13 ± 7.47	374.53 ± 7.50
VS (g kg^{-1})	854.14 ± 4.80	557.33 ± 20.79	219.67 ± 3.57	179.72 ± 7.88	108.68 ± 4.35	521.52 ± 7.70	366.94 ± 7.23
TKN (g kg^{-1})	3.21 ± 0.54	13.80 ± 0.36	26.32 ± 10.30	12.38 ± 0.69	1.32 ± 0.18	10.75 ± 0.11	11.69 ± 0.11
NH_4^+–N (mg kg^{-1})	148.51 ± 2.06	389.70 ± 64.67	1512.37 ± 19.53	1645.81 ± 15.00	60.83 ± 2.05	274.51 ± 77.64	537.55 ± 195.10
$Protein_{estimated}$ (g kg^{-1})	*19.14*	*83.80*	*155.04*	*67.10*	*7.88*	*65.48*	*69.73*
Fat_{soxtec} (g kg^{-1})	762.79 ± 2.39	467.30 ± 14.90	46.60 ± 1.50	86.64 ± 0.01	18.42 ± 1.18	432.07 ± 4.80	249.38 ± 2.45
COD (g kg^{-1})	n.d	n.d	n.d	n.d	n.d	1356.03 ± 21.66	849.24 ± 77.99
$COD_{estimated}$ (g kg^{-1})	*2294.47*	*1473.46*	*385.40*	*376.85*	*152.01*	*1363.66*	*871.01*
Liquid waste and final mixtures (M)	**Waste blood**	**Screening waste**	**DAF sludge**		**Wastewater**	**M1**	**M2**
TS (g kg^{-1})	196.60 ± 0.01	242.30 ± 2.72	95.65 ± 0.78		0.93 ± 0.02	79.64	36.00
VS (g kg^{-1})	184.04 ± 0.03	233.34 ± 2.61	82.66 ± 0.76		0.53 ± 0.03	77.94	34.36
TKN (g kg^{-1})	31.70 ± 3.01	12.40 ± 0.32	5.87 ± 1.52		0.15 ± 0.01	1.89	1.67
NH_4^+–N (mg kg^{-1})	133.00 ± 9.90	1502.21 ± 20.14	861.00 ± 9.90		84.00 ± 2.30	149.09	135.17
$Protein_{estimated}$ (g kg^{-1})	*197.34*	*68.12*	*31.29*		*0.41*	10.88	9.59
Fat_{soxtec} (g kg^{-1})	n.d	107.11 ± 0.64	50.30 ± 0.05		0.10 ± 0.01	63.29	20.92
$COD_{estimated}$ (g kg^{-1})	*262.73*	*471.18*	*192.44*		*0.92*	*201.96*	*78.51*

[85] is technical report based on 15 years' experience of full-scale operational AD of SHW in Linköping, Sweden, and is the only available such report found in-keeping with the stated

research constraints (scoring high on technological representativeness, accuracy, precision and data quality, although it scored lower for temporal representativeness). It provides an exhaustive analysis of process limitations and recommendations for optimal operating conditions, and despite its assertion of the challenges associated with processing SHW compared to traditional feedstocks, it concludes that its mitigation measures (evaluated upon in section 6) have led to an increase in total biogas production per unit mass of feedstock over its 15-year operational period over which it has produced upgraded vehicle-grade biomethane.

Challenges cited include an increased ammonia production from fermentation of proteins and fats found in high concentration as described which leads to a pH increase acting as a growth limiting agent to volatile fatty acid (VFA) consuming methanogens. Additionally, the degradation of protein-rich substrates leads to the formation of sulphides which, in addition to having a corrosive effect on plant equipment, also result in sulphide inhibition of the methanogens [85]. [63] provides a temporally, technologically, and geographically representative study of the AD of cattle SHW sub-streams from Ireland. Although the SHW composition and characterisation is identified through analysis of primary data (providing an accurate representation of the categorisation and methods of producing waste in SH facilities), the AD process was simulated using biochemical methane assays, and the theoretical methane yield was calculated from the elemental constitution of the experimental product streams via the Buswell equation (below), therefore scoring lower on accuracy.

$$C_cH_hO_o + \left(c - \frac{h}{4} - \frac{o}{2}\right)H_2O \rightarrow \left(\frac{c}{2} + \frac{h}{8} - \frac{o}{4}\right)CH_4 + \left(\frac{c}{2} - \frac{h}{8} + \frac{o}{4}\right)CO_2$$

Figure 4: Buswell equation for the calculation of theoretical methane yield

The sub-streams were categorised as such: paunch, dissolved air flotation sludge, soft offal (SO) and slaughterhouse waste mixed at annual production levels (SHWM). Its aim was to prove the viability of anaerobically digesting cattle SHW sub-streams and evaluate its impacts on energy recovery in bioeconomic systems in Ireland.

It proved a high methane yield could be achieved (641.55 mLCH4 g/VS), and its net energy analysis showed that the recovered energy had the potential to subsidise 2.07% and 1.38% of the industrial sectors final electrical and thermal demands in Ireland in 2013 [63, p. 547].

In assessing NI's capacity to provide SHW feedstock for AD, [75] provides an exhaustive quantification of AD feedstocks within the contacts of the NI bioeconomy, scoring high for geographical, temporal and technological representativeness), although its analysis is based upon a theoretical estimation formulated via a secondary methodology, scoring lower for accuracy and precision. Nevertheless, it presents the best available indication of the current regional landscape. It found that SWH ranked near the top of its defined waste categories for overall quantity produced (t/year). [59] outlines the comparatively high stocking density of animal feed feedstocks in NI and provides an indication as to the animal type that may provide most economic benefit to the region (as regards to crude livestock stocking density per hectare).

Table 5: Livestock numbers by type (UK) [59]

	England	**Scotland**	**Wales**	**NI**
Total livestock numbers	Cattle – 5,146,000 Sheep- 15,390,000 Pigs – 3,776,000 Poultry – 138,850,000	Cattle – 1,730,000 Sheep – 6,670,000 Pigs – 319,300 Poultry - 14,900,000	Cattle – 1,137, 399 Sheep – 10,037,474 Pigs – 24,503 Poultry - 7,515,804	Cattle – 1,611,776 Sheep – 1,986,932 Pigs – 674,428 Poultry – 24,780,381
Approximate crude livestock stocking densities per hectare (livestock nos ÷ by total farmed area)	Cattle – 0.56 Sheep – 1.6 Pigs – 0.41 Poultry – 15.15	Cattle – 0.30 Sheep – 1.17 Pigs – 0.05 Poultry – 2.63	Cattle – 0.71 Sheep – 6.29 Pigs – 0.01 Poultry – 4.7	Cattle – 1.57 Sheep – 1.94 Pigs – 0.65 Poultry – 24.21

Through a comparison of crude livestock stocking densities per hectare, it was found that poultry offers the largest opportunity for the investment of AD technologies, which is congruent with findings in [86] that advocates for the utilisation of poultry biomass in AD through the evaluation of the methane yields of various animal types as SHW feedstocks, as well as their degradation efficiency (poultry ranks among the highest on both counts). [86] is a lab-based study of the AD of SHW conducted in Sweden, scoring low on technological,

geographical, and temporal representativeness, as well as accuracy, but very high on precision.

Table 6: Biodegradation of different animal wastes in anaerobic mono-digestion [86]

Substrates	T(°C)	Operation Mode and conditions	Methane yield (Y_{CH_4})	Degradation efficiency (%)
Slaughterhouse waste				
Solid cattle	55	Batch (2L)	582-681 (mL gVS^{-1})	n.a
	55	CSTR (3L), HRT= 25 days, OLR=0.9 gVS $L^{-1}d^{-1}$	410 (mL gVS^{-1})	n.a
Solid cattle and pig	35	CSTR (2L), HRT= 30 days	60 (mL gVS^{-1})	34.5 (VS red)
Solid and liquid pig/cattle mixtures	35	Batch (1L)	274-302 (mL $gCOD^{-1}$)	86-92 (COD red)
Iberian pig	38	CSRT (2L), HRT=23.5 days	17.84 (m^3 m^{-3}_{Feed})	78.59 (COD red)
Pig	55	Batch (0.5 and 2L)	230-620 (mL gVS^{-1})	50-100 ($Y_{CH_4}/Y_{CH_{4Theoretical}}$)
	35	Batch (1.2L)	460 (mL gVS^{-1})	76.6 (COD red)
Poultry	35	Batch (1.2L)	580 (mL gVS^{-1})	55.2 (COD red)
	35	CSTR (2L), HRT= 50 days, OLR=0.8 gVS $L^{-1}d^{-1}$	520-550 (mL gVS^{-1})	n.a
	34	CSTR (3L), HRT= 50, 36, 25 days, OLR=0.9, 1.16, 1.70 gVS $L^{-1}d^{-1}$	600-700 (mL gVS^{-1})	77-79 (VS red)

Accordingly, poultry feedstock provides the largest opportunity for investment into waste-management technologies. However, attempting to quantify SHW facilities in NI and associated waste stream compositions/biogas yields proved difficult, so the geographical constraints on the search criteria were widened to encompass a global reach (representing an area for further research development).

[87] is a lab-based study (scoring lower on accuracy) into the AD of poultry SHW in Brazil, offering both technical considerations for the specific treatment of poultry waste (scoring high on technological representativeness), and a detailed characterisation of poultry SH effluent streams and resultant biogas composition and methane yields. Despite presenting a complete and relevant date source, it scored low on temporal and geographical representativeness.

It was thought that the substitution of cattle SHW for poultry SHW would prove more applicable to this study due to the perceived increased prevalence of regional primary data referring to the characterisation of cattle effluent streams and resultant methane yields from AD, but this method did not yield workable results. 17 abattoirs in the vicinity of the AFBI plant were contacted for data and only 1 responded in kind (Lakeview Farm Meats, although

it was found that they do not produce enough waste per day to facilitate the operation of the AFBI CSTR capacity). There is no available evidence of operational AD plants utilising poultry SHW in NI, nor of operational characterisations of poultry SWH streams.

[88], however, represents a lab-based study of the AD of feedstock from a real poultry SH in Cork, Ireland conduction in 2016 via biochemical methane potential assays (meeting the criteria for the standard of both temporal and geographical representativeness). It also comprehensive, provided peer-reviewed data for both the characterisation of the feedstock from the SH and the resultant biogas composition (including methane yield). Furthermore, it provided data based on operational SH sub-streams and retention times (congruent with the AFBI technical report) as shown below, providing a more practicable assessment of the viability of this feedstock.

Table 7: Characteristics of poultry waste streams. [88]

	DAF	SO	PSO	SHWM
TS[a]	11.7 (0.59)	40.3 (3.69)	44.9 (2.17)	24.2 (0.21)
VS[a]	93.4 (0.02)	96.6 (0.70)	96.6 (0.70)	94.8 (0.11)
Fat[b]	2.7	36.1	36.1	24.3
Protein[b]	53.5	34.0	34.0	40.9
Carb[b]	37.2	26.5	26.5	29.6
C[b]	51.6	55.3	55.3	54.0
H[b]	7.8	9.4	9.4	8.9
O[b]	25.2	26.7	26.7	25.5
N[b]	8.8	5.1	5.1	6.4

a % (standard deviation).
b % of TS.

Table 8: Results of BMP assays [88]

Substrate	IP[a] (days)	V_0 (mlCH$_4$ gVS^{-1})	$V_{0d\ max}$[b] (mlCH$_4$ gVS^{-1} d^{-1})	80% of V_0 (days)
DAF	30	261.35 ± 20.39	46.19 ± 1.34 (1)	10
SO	30	499.11 ± 35.44	32.80 ± 0.39 (17)	21
PSO	50	501.13 ± 40.59	45.81 ± 0.14 (1)	36
SHWM	50	594.59 ± 38.05	45.10 ± 1.42 (23)	32

Note: relative standard deviation applied where necessary.
[a] IP- Incubation period of BMP.
[b] ( ) day on which maximum daily V_0 occurred.

Therefore, [88] was selected for inclusion in the model, utilising the data inputs above (the PSO sub-stream was selected due to its ranking as having the highest TS and presenting as a more homogenous stream than SHWM.

A rudimentary search on google maps was undertaken to identify suitable poultry feedstock sources based on distance from the AFBI site (in accordance with findings in [36] that feedstock transportation represents the highest GHG emission burden within the AD refinery), the results of which are shown below.

Table 9: Distances of Slaughterhouse Waste Sources from AFBI Hillsborough Site

Feedstock Source Site	Distance from AD Facility (AFBI Hillsborough Plant)/km	Feedstock Type
Moy Park, Dungannon	52.6	Poultry Waste
Moy Park, Ballymena	55.4	Poultry Waste
Moy Park, Craigavon	32.1	Poultry Waste

N.B. the distance as it appears in the model is calculated in miles (32.1km = 20 miles)

The selected site was Moy Park Craigavon, which produces quantities of poultry feedstock more than the tonnage demonstrated by the Hillsborough site capacity as utilised in this study's model [89].

The feedstock will be transported to the AD refinery by HGV trucks on-road in a weekly delivery.

3.2.2. AD Plant

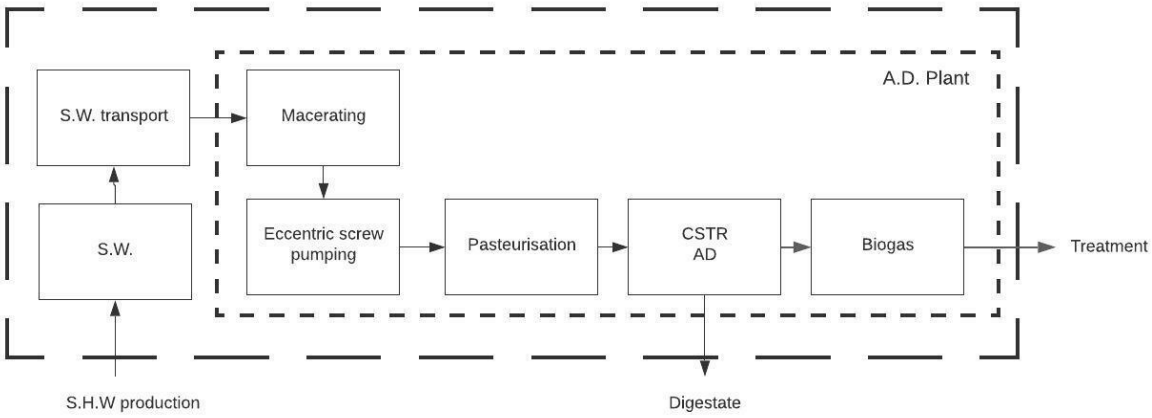

Figure 5: System boundary for the A.D. plant

Pre-treatment

Initial search keywords: anaerobic digestion, pre-treatment, macerator(ing), grinder(ing), pasteurisation(er), biogas production, biorefinery

Sources assessed against traffic lighting criteria: 7

[63] was selected from the literature review for inclusion in the model: the study was conducted in Ireland and is widely cited (41 citations in Scopus with a Field-Weighted Citation Impact of 1.15 (cited more than expected)). A limitation of the model is that the data is estimated using theoretical energy model on a cattle slaughterhouse facility in Cork, Ireland, and not based on an existing operating AD plant processing poultry SHW. However, no technical reports were discovered on an operational AD plant in Ireland or the rest of Europe that processes unmixed SHW, therefore the study represented the best available source for inclusion in the model.

The following information from [63] was used in the pre-treatment LCA model:

- The macerator which is a heavy-duty grinder used to reduce heterogeneity by decreasing the particle size (< 8mm) of the organic material to create a homogenous feed that is pumpable. The unit had an energy requirement of 6.25 kWh/tonne which in agreement with other macerator units of comparable size. [90] [91] [92]

- The eccentric screw press pump which was used to pump the thick feed between units. The unit had a similar energy requirement to the macerator of 6.25 kWh/tonne which is in consensus to other industrial units found on the market. [93] [94]
- The pasteurisation unit with operating conditions of 70°C for 60 minutes. The electric power consumption was estimated by calculating the energy requirement needed to raise the temperature of water inside the SHW from ambient conditions to 70°C and multiplying this temperature difference by the specific heat capacity of water. This method gives an acceptable estimate, due to the high moisture content in SHW (~75%) [95].

The study is based around the pre-treatment for cattle SHW whilst poultry SHW was used as the feedstock in the LCA model. Notwithstanding, the same pre-treatment method can be utilised with the same estimated energy requirement because cattle and poultry carcasses have similar high composition of fat and protein (Cattle 58% fat and 26.5% protein: Poultry 41.8% fat and 31.6% protein) [96]. Pasteurisation in itself is an energy intensive process but increases the methane yield due to improved bio-kinetics (Poultry Specific Methane Yield (SMY), unpasteurised 465.34 $mLCH_4\ gVS^{-1}$, pasteurised 518.18 $mLCH_4\ gVS^{-1}$) [96]. The energy requirement is affected by seasonal variation as the lower the ambient temperature of the feed the more energy is required to raise the temperature. In Hillsborough the average temperature during the winter months (December – January) is 4.8°C in comparison to the summer months (June – August) with an average temperature of 14.1°C. [97]

Using an eccentric screw pump increases the electrical power requirements of the pre-treatment further but the pump's ability to operate under constant pressure to produce steady state flow between different units means it can transport heavy viscous liquids. [63] An alternative method of operation was seen in [85] which is a full-scale operational AD plant in Linköping, Sweden, utilising co-digestion of SHW with liquid slurry. The proportion of SHW in feed mixture varied between 35% and 75% and was homogenised in the macerator unit to produce a less viscous feed for easier pumpability resulting in a smaller needed energy requirement. [85] After the feed was thermally heated in a buffer tank to 75°C to limit foaming, which is formed by the production of inhibitory compounds otherwise known as intermediate fermenters such as Volatile Fatty Acids (VFAs) and Long Chain Fatty Acids (LCFAs) along with protein as a bio-surfactant. [85] Foaming lowers the pH of the digester which in return lowers the methane yield. [85] However, [98] argues the extra pre-thermal

treatment step adds little value to the overall pre-treatment process and the pasteurisation unit provides satisfactory thermal treatment to prevent foaming problems in AD unit.

The technical reports from the two operational AD plants reviewed described a split feed wherein the feed was processed in 2 streams and 2 separate pasteurisation units, allowing for improved process continuation of the process. [85] [99]

Digester

Initial search keywords: anaerobic digestion, AD, CSTR, PFR, digester unit, slaughterhouse waste, food waste, FW, SHW

Sources selected for traffic light matrix assessment: 6

As described in section 1, AD occurs as a result of 4 sequential processes – hydrolysis, acidogenesis, acetogenesis and methanogenesis. During the hydrolysis stage, insoluble organic matter and compounds with higher molecular mass are transformed into soluble organics that can be easier utilised by acidogenic microorganisms in the next stage. Acidogenic microorganisms degrade the smaller monomers created during hydrolysis into organic acids, alcohols, hydrogen, and CO_2. During the acetogenic phase, the compounds produced in the previous stage are then converted to methanogenic substrates [100]. Note that many products of acidogenesis can be utilised directly for methanogenesis, however acetogenesis is key in maximising conversion of all substrates and therefore maximising biogas yield [101]. Finally, during methanogenesis, acetate is broken down into CH_4 and CO_2 while CO_2 and hydrogen combine to form CH_4, requiring strict anaerobic conditions.

In selecting a digestion configuration, the following factors were considered:

- reactor type [102]
- mixing [103]
- reactor conditions (mesophilic vs thermophilic) [104].
- dry base/wet based digestion (dependent on the water content of the feed [52])

The most common reactor types identified through the literature review were CSTR, PFR, batch-fed reactors, and membrane reactors [105]. Reactor conditions also have a large impact on the yield of biogas from AD systems. Mesophilic systems operate at a lower temperature range of 30-45°C encouraging the growth of mesophiles, a family of microorganisms which

thrive in this temperature range. Thermophilic systems operate between 45-60°C, promoting the growth of thermophiles. Mesophilic reactors require less process energy due to the lower temperature range and are more frequently used in practise due to their higher stability [106]. While thermophilic systems often provide a larger yield of biogas [107, 108] and lower hydraulic retention times [106], the studies considered for relevant analysis presented narrow, lab-based using small-volume reactors (upon scale-up it is frequently found that thermophilic systems are unstable and susceptible to temperature shock [109], requiring frequent monitoring and maintenance). Therefore, mesophilic conditions are preferred.

The literature selection process identified 17 papers which met the research criteria, and of these, six were found that fit traffic light criteria. [105] reviews common reactor types for small-scale AD systems on farms across Europe. Of these, it was found that CSTRs were most common and among the most effective for production of biogas from a range of animal wastes. [110] analysed small-scale AD systems on Irish farms, with CSTRs used exclusively in the sample reported. Finally, [67] is a technical operation report on the AFBI plant. This source was considered most relevant to the modelling of this study's baseline scenario, and efforts were made throughout this study to maintain process conditions as close as practicably possible to those of the AFBI plant. As such, the following process conditions were chosen for the model development (congruent with the AFBI technical report): utilisation of a CSTR with a volume of 660m^3 and a hydraulic retention time of 30 days.

In stipulating reactor conditions for the model, [67, 109, 111] were analysed to determine the optimum temperature and mode of operation, and these conditions were compared against those of the AFBI technical report. [111] describes the undertaking of an LCA of a grass-fed anaerobic digestion plant in Ireland considering dry and wet feeds. While it found dry fed reactors operate well under thermophilic conditions at 50°C, wet fed reactors (like the plant considered in this LCA) operate better under mesophilic conditions at 38-40°C and are more stable and less susceptible to temperature shock. [109] investigated thermophilic and mesophilic systems to determine which are more effective for large-scale operation. Like the previous papers, it found that thermophilic conditions provide a better biogas yield however the increased heating costs and process instability leaves mesophilic as the preferred mode of operation for large-scale AD systems. Conventionally, thermophilic systems are preferred for the elimination of pathogens in digester substrates [68] however as the feed to the digester was pasteurised, a lower reaction temperature was acceptable. Accordingly, it was decided that the reactor would operate under mesophilic conditions at 39°C. This is congruent with

conditions stated in [63] for the lab-based experimental analysis of methane yield from cattle SHW in Ireland (refer to section 3.2.1). Vigorous mixing is required to avoid the formation of an impenetrable crust as illustrated in [69]. For the mixing of the reactor contents, [67] provides the basis for the utilisation of gas recirculation as an effective, low-energy method of homogenising substrates inside the reactor. As the solids content within the digester was low, pneumatic mixing is a viable alternative to mechanical stirring [68]. Gas recirculation only requires activation for a few minutes every hour [67], reducing overall energy requirements. Furthermore, biogas recirculation has been found to improve digester performance and increase biogas yield [112]. Compared to mechanical stirring, it is found that prolonged periods of mixing do not improve performance [113], rather it can reduce the overall electrical efficiency of the digester [114]. For these reasons, biogas recirculation was chosen as the mixing method for the digester.

3.2.3. Scenario 1: CHP

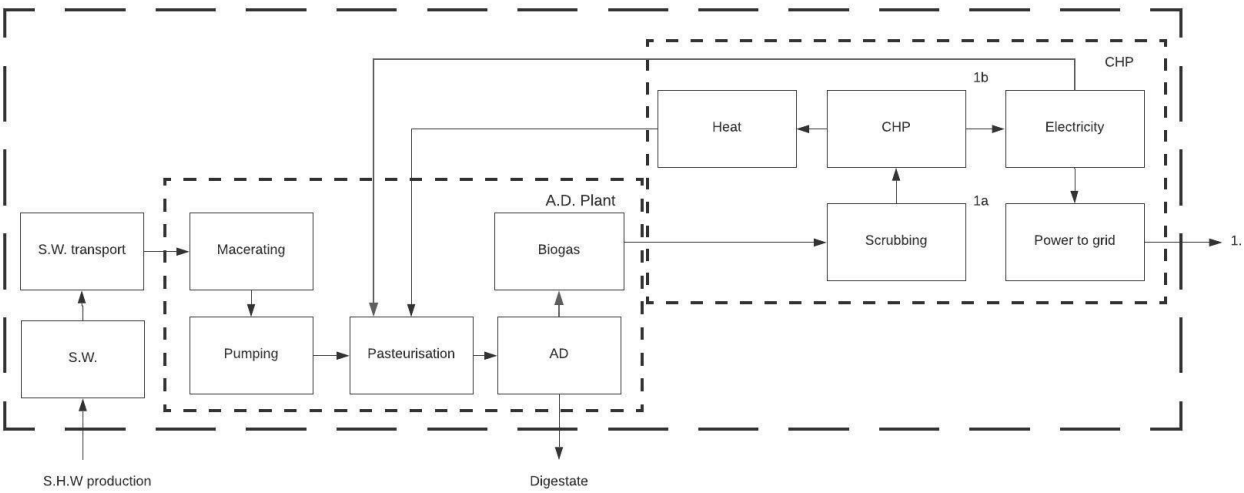

Figure 6: System boundary for scenario 1 (CHP, power to grid)

H_2S Removal

Initial search keywords: anaerobic digestion, AD, scrubbing, H_2S, H_2S removal, biogas cleaning

Sources selected for traffic light matrix assessment: 4

The literature review resulted in the consideration of the following H_2S removal methods: biological desulphurisation [115], iron chloride dosing [116], bio-trickling filtration [78] and activated carbon [67]. Biological desulphurisation involves introducing a small, controlled volume of O_2 in the reactor headspace, promoting growth of a species of microorganism that feeds on sulphur-containing compounds in the feed [115]. Iron chloride dosing is the addition of iron chloride to the digester, resulting in a reaction where Fe3+ is reduced to Fe2+ before reaction with sulphur to form F2S3, a non-soluble compound that precipitates out of the substrate in the digester [116]. There are some limitations with this method, namely the effect on pH and nitrogen concentrations inside the digester [117]. Bio-trickling filtration makes use of a packed column with a support containing sulphide-oxidising bacteria [78]. While trials have been carried out on an industrial scale, issues are cited relating to elemental sulphur production clogging the column [118]. There limitations of this research were mostly attributable to the deficiency in sources relevant to technology used in Ireland. Many of the techniques researched were small-scale or laboratory-based studies and data pertaining to large-scale plants proved difficult to obtain. For these reasons, the most relevant study was [67], the AFBI Hillsborough technical report which makes use of an activated carbon column for desulphurisation of biogas. As a technical report based on an AD plant in NI, it has significantly better geographical and technological relevance compared to other papers found throughout research on this topic. A small amount of energy was required (approximately 12 kw) to increase the pressure of the biogas to 2-7 bar before it entered the column [119].

CHP Unit

Initial search keywords: CHP, combined heat and power, anaerobic digestion, AD, biogas engine, biogas production

N.B. the traffic light matrix was not utilised in this section, as a combination of sources were essential in fulfilling the full requirements of the unit operating conditions. However, the criteria outlined in the traffic light matrix was referred to throughout the design selection process, and the same constraints outlined in section 3.1 were applied.

CHP units generally consist of an engine wherein biogas undergoes combustion to generate heat and electricity and a heat exchange system to recapture excess heat. A key advantage of

CHP units is the ability for a plant to be self-sufficient provided their biogas output is high enough to offset the plant power requirements [120]. On average, CHP units have electrical efficiencies ranging from 30-40% and thermal efficiencies ranging from 35-55% [121], offering potential overall efficiencies of up to 95%. Compared to some gas turbines with efficiencies between 30-50% [122], CHP is an attractive alternative provided the heat energy produced can be utilised effectively.

In terms of reducing GHG emissions, CHP is a good option offering reductions of up to 30% [123]. Biogas utilisation in CHP units is considered carbon-neutral, however comparison against other CHP fuels can be made in terms of global warming potential (GWP). Compared to natural gas CHP units biogas offers GWP reductions of 34%, while utilising biogas in lieu of fossil fuel based CHP units can reduce GWP by up to 50% [124]. As a by-product of AD processes, H_2S must be removed before biogas can be used in CHP units. Compared to coal-fired stations this provides a large reduction in SO_2 emissions. In terms of reducing GHG emissions, CHP is a good option offering reductions of up to 30% [123]. Biogas utilisation in CHP units is considered carbon-neutral, however comparison against other CHP fuels can be made in terms of global warming potential (GWP). Compared to natural gas CHP units, biogas offers GWP reductions of 34% while utilising biogas in lieu of fossil fuel based CHP units can reduce GWP by up to 50% [124]. As a by-product of AD processes, H_2S must be removed before biogas can be used in CHP units. Compared to coal-fired stations this provides a large reduction in SO_2 emissions.

On a local scale, CHP can be used to help NI achieve a lower carbon economy. As the largest source of GHG emissions in NI is through the agriculture sector [125], utilising small-scale CHP plants across a number of farms can help offset emissions. Producing heat and electricity for farm equipment and buildings through CHP opposed to from the grid will reduce the impact the sector has on the environment overall. However with a capital cost averaging £5000 for a micro CHP boiler in NI alone [126] without consideration for the cost of the rest of a small-scale AD plant, it is not economically feasible for most farms to install the system without government subsidies. Furthermore, following the closure of ROCs for AD plants by the UK government in 2017 [127], it is unlikely that farmers or investors will be encouraged to pursue AD-based CHP systems.

As previously identified, CHP is the most common utilisation pathway for biogas produced from AD in Ireland. owing in part to the lack of infrastructure for biomethane injection to the

natural gas grid [128] (refer to section 3.2.4), giving rise to a large pool of high quality primary data on the efficiency of CHP. A technical specification document from a manufacturer [129] was used to determine the electrical and thermal efficiencies utilised in this LCA (33% and 44% respectively).

3.2.4. Scenario 2: G-t-g

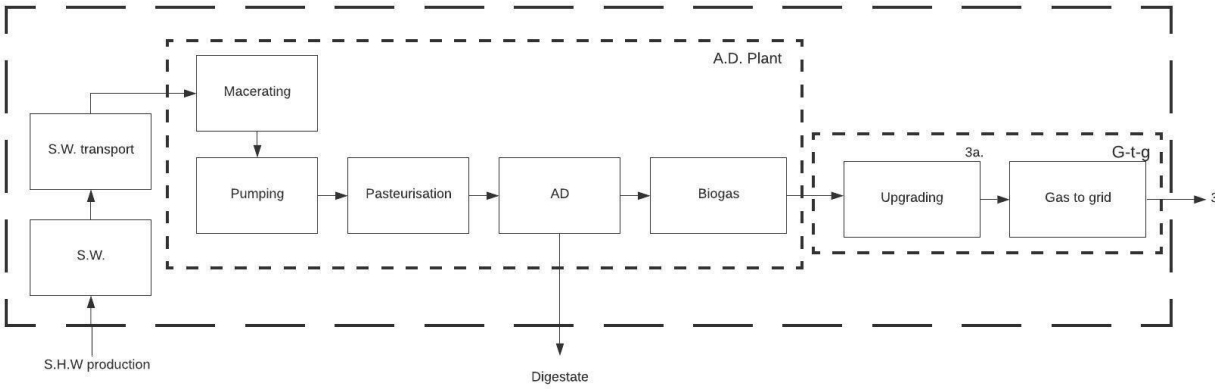

Figure 7: System boundary for scenario 2 (G-t-g)

Initial search keywords: anaerobic digestion, AD, biogas, upgrading, PSA, scrubbing, biomethane, biomethane production, injection to the grid, RNG

Sources selected for traffic light matrix assessment: 7

The purpose of upgrading biogas to biomethane is to reduce the carbon dioxide (CO_2) concentration to increase the proportion of methane (CH_4) (> 96%) [130] and replicate the energy density of natural gas. Biomethane generated AD is regulated by the Quality Protocol [131], developed by Waste and Resources Action Plan (WRAP) and other stakeholders. If the biomethane meets the criteria outlined it is no longer regarded as a waste and may be injected into the national grid for space heating. [131] This regulation is in use in the UK (England and Wales) but not NI as there is no infrastructure currently available to facilitate biomethane injection into the grid rendering the protocol not applicable. If, however, investment into relevant infrastructure is facilitated, the same criteria would most likely apply and is therefore

adopted in this study. Biomethane must additionally meet NI entry specification for combustion energy output, defined by the Wobbe Index (WI), at 45.7 MJ/m^3 [132]. Legal requirements vary by country for WI in the national grid. [133] In Denmark, which widely uses biomethane in the national grid [134], raw biogas from a manure feed AD plant was upgraded via water scrubber with an increase in WI from 28.3MJ/m^3 to 51.3MJ/m^3 [135] which exceeds the national NI grid criteria [132]. The initial WI of raw biogas is dependent on feedstock [136], since slaughterhouse waste has a similar methane concentration in the biogas as manure produced biogas [112] which should produce similar value of WI. Overall to meet the national gas grid specification, the biomethane needs to meet the following specification: CH_4 (> 96%), CO_2 (< 2%), H_2S (5 mg/m^3), O_2 (0.2 < 0.5%) and 3 – 20 mg/m^3) [130].

[53] undertakes a systematic review of commercially viable upgrading technologies, identifying five available physical/chemical separation technologies (other methods based on cryogenic processing and chemical hydrogenation were also discussed, but dismissed from the analysis due to not meeting the threshold of commercial viability). These separating methods operate by different working principles to achieve the desired biomethane quality as described below [53]:

- Physical absorption method can be used with either water or an organic solvent which relies on that CO_2 is more soluble in water/organic solvent in comparison to CH4 in accordance with the Henry's Law.
- Chemical separating techniques uses aqueous amine solutions to bind to the CO_2 molecules inside the biogas.
- Pressure swing adsorption (PSA) operates by separating the gasses inside the biogas based on the molecular characteristics and affinity of the adsorbent materials.
- Membrane technology uses the selective permeability properties associated with membranes to separate the components.
- Cryogenic separation process operates by decreasing the temperature of the biogas to allow CH4 to liquify from the CO_2 and the other biogas components.
- Chemical hydrogenation uses a catalyst commonly with Nickel and Ruthenium under elevated temperature conditions to convert CO_2 and H2 to CH4 with high selectivity.

These methods were evaluated and compared against each other in terms power consumption, percentage lost and recovered of CH_4, pre-purification steps needed, ability to remove H_2S, operation pressure and outlet pressure.

It was found that physical absorption had relative high loss of methane (4 % CH_4 loss), cryogenic had a high energy consumption (0.76 kWh/Nm^3), chemical absorption required further pre-purification and could not filter out H_2S. Whilst water scrubbing and PSA was able to filter out H_2S and needed relative low consumption of energy (PSA 0.23 kWh/Nm^3, Water scrubber kWh/Nm^3). Water scrubber had smaller loss of methane than PSA (PSA < 4 % CH_4 loss, Water scrubber < 2 % CH_4 loss) but performed worse than membrane separation (< 0.6 % CH_4 loss). Apart from chemical absorption with a higher methane recovery (96 - 99% CH_4 recovered) all the units had a similar methane recovery (96 – 98% CH_4 recovered). In AD industry, water scrubber is the most widely used but other units such as PSA is more prominently used in low scale industry.

As mentioned previously the feed leaving the CSTR contains H_2S which needs to be filtered out, low energy consumption with high methane recovery and low methane loss are other desirable prosperities to meet the NI national gas composition criteria. Therefore, water scrubber and PSA were both seen as attractive upgrading options. PSA had an outlet pressure of 4 – 5 bar whilst the water scrubber had an outlet pressure of 7 – 10 bar. According to the pressure classification of Northern Gas Networks, PSA is categorised an intermediate pressure (7 bar > 2 bar), whilst water scrubber as a high pressure (> 7 bar). [137] Overall, the water scrubber is able to accommodate high volumes according to [53] review with its view supported by other studies and technical information whilst simultaneously meeting all the criteria of the NI grid gas specification and therefore selected as the upgrading technology for the LCA model. In conjunction with the traffic light system, [53] was selected to base the model on due to its high temporal representation and similar technology existing in NI.

The residential sector is responsible for 14% of NI total GHG emissions with 2.6 MtCO2e released in 2018 [138], making it the third largest GHG emitter sector, but simultaneously considered one of three sectors where there has been the largest decrease in GHG emissions [139]. This is largely due to improvements in domestic energy efficiency and fuel switching (coal to natural gas) [138]. NI domestic heating are mostly fuelled by oil (68% households), followed by gas (28% households) with solid fuel and electric compromised the rest (8% households) [140]. Most of the gas lines are in urban settings with continuous expansion of

lines, with 232,920 connected in Greater Belfast Area via Phoenix Natural Gas (PNG) [141], one of NI three gas distribution network companies. The other two gas distribution companies are Firmus Energy and SGN Natural Gas Limited who are rolling out gas distribution lines to North-West (Ballymena, Antrim, Banbridge, Craigavon, Portadown Armagh and Newry). [121]. However, presently there is poor converge in rural areas where most AD plants are located. [122] Biomethane can be injected into high pressure gas transmission grid, or the local low pressure gas distribution grid as a substitute or in conjunction with natural gas. By injecting biomethane into the grid the proportion of fuel generated via renewable route would increase whilst simultaneously reduce fossil fuel dependence. Natural gas produces 666 gCO_2/MJ [142] burnt whilst CO_2 emissions generated from biomethane is dependent on feedstock, wet manure, maize and biowaste produced -79 gCO_2/MJ, 50 gCO_2/MJ and 35 gCO_2/MJ respectively for closed digestate [143], suggesting a great CO_2 saving from producing biomethane from slaughterhouse waste and subsequent injection into the national grid for space heating.

In NI the rate of fuel poverty is 22 per cent considering three factors: income, cost of energy and domestic energy efficiency at home resulting in the household spending more than 10% of its income on energy costs. [144] NI is dependent on other countries for natural gas and the cost is affected by external factors such as amount produced, level storage and variations of seasons [145]. By producing biomethane via AD creates a more secure supply of fuel for NI, although infrastructural barriers currently present a challenge for this technology and warrants further discussion (only 103 biomethane-to-grid biogas plants are in operation across the UK [56]).

3.2.5. Scenario 3: HGV Fuel

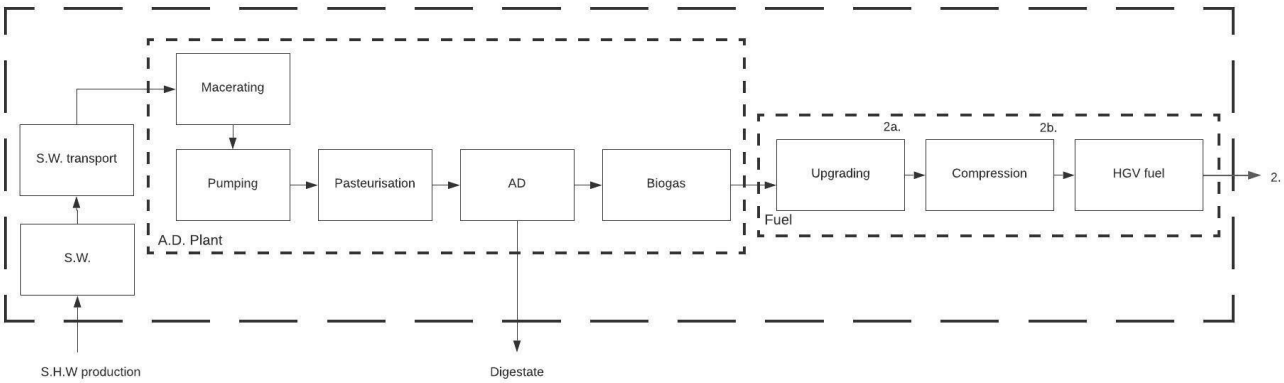

Figure 8: System boundary for scenario 3 (HGV Fuel)

Initial search keywords: HGV fuel, compression(er), upgrading, anaerobic digestion(er), AD, LNG, biodiesel, biomethane

Sources selected for traffic light matrix assessment: 6

Transport is the second largest net contributor to GHG emissions in NI behind agriculture [125]. Emissions caused by transport have increased by 22% in the period between 1990-2019 [125] therefore it is imperative that these are reduced, either through decarbonising fuel sources or reducing the extent of transport. However, unlike agriculture which society is dependent upon for food, there is far larger scope for the reduction of these emissions through the utilisation of alternative fuels, such as electricity or biofuels [146]. Over their lifetime, biofuels can be considered carbon neutral, as any carbon emitted during combustion is assumed to be sequestered back into a later batch of crops through photosynthesis [147]. Compressed biomethane is a strong contender for biofuel, with a calorific value of 36-39 MJ/m^3 [148]. Compared to other potential sources of biofuel such as rapeseed oil (33.76 MJ/m^3) or castor oil (33.97 MJ/m^3) [149], biomethane has a greater energy density. Furthermore, there is contention around using food crops and arable land as a basis for bioenergy, with some arguing that as populations and demands for food and biofuels grow, alternative sources to food crops must be utilised [150]. As biomethane can be produced from anaerobic digestion of waste feedstocks, there is no need to sacrifice land and food sources for energy crops. In NI, there is already

precedent for the utilisation of biomethane as HGV fuel. McCulla LTD, one of NI's largest transport companies, partnered with WELTEC Biopower in 2017 to construct a biogas plant. Originally biogas was utilised through a 500kW CHP unit however recently the company has extended its interest into bioenergy, replacing the diesel fuel of 10 of its HGVs with biomethane [151]. Additionally, Circle K's Dublin port public fast-fill CNG station has the capacity for refuelling 70 HGVs a day [152]. However, due to poor infrastructure and uptake, there are only 7 compressed natural gas stations currently in operation in Ireland, with a further 7 public stations in planning and development. Currently the majority of stations in Ireland are privately owned, limiting the availability of biomethane as an alternative fuel [152].

There are substantial reductions in GHG emissions in substituting diesel for biofuels. An estimated 5.09 kgCO_2/km travelled could be saved by HGVs switching to biofuels [153]. For an annual distance of 50000km, this would remove 254.5 tonnes of CO_2 from the atmosphere per HGV per year [153]. Another estimate of 65% CO_2 reductions when comparing compressed biomethane from AD to diesel in HGVs further reinforces biofuels as an attractive option [154]. Furthermore with an estimated price of 13.8p/km, similar to that of diesel and resulting in a CO_2 savings cost of £1/tonne CO_2 [154], biomethane can be considered an economically feasible alternative to conventional fuels. To meet the commitments laid out in the Paris Agreement [5] it is not simply enough to reduce the production and use of fossil fuels, rather remaining reserves of coal, oil and methane gas reserves must remain unexplored [155]. To this end, expansion of biofuels and electric vehicles must become a priority over the coming years [156].

Biomethane must be compressed to 200-250 bar [80] for utilisation as HGV fuel, requiring a compressor duty of 0.35kWh/m^3 biogas [111].

4. Methodology

4.1. Inventory Compilation

The input and output data utilised for modelling this LCA is tabulated below, with energy requirements and GHG emissions quantified in terms of the chosen FU, tonnes of feed.

Table 10: Input data for general AD Plant

Data	Value	Unit	Source
Input to AD Plant			
SHW	7300	tonne per year	Calculated
Macerator	8.14	kWh/tonne feed	Ware and Power (2016) [63]
Pasteurisation Energy	31.81	kWh/tonne feed	Calculated
CSTR Heating	32.00	kWh/tonne feed	Frost et al (2011) - AFBI Plant [67]
Gas Recirculation Energy	3.58	kWh/tonne feed	Mahmoodi et al (2020) [157]
SHW Methane Yield	499.11	m^3/tonne VS	Ware and Power (2016) [63]
Output of AD Plant			
Biogas	271.57	m^3/tonne feed	Calculated
Digestate	0.63310	tonne/tonne feed	Calculated

Table 11: GHG Inventory for Scenario 1: CHP

Data	Value	Unit	Source
Scenario 1 - Electricity to Grid from CHP Unit			
H_2S Removal Energy	13.02	kWh/tonne feed	Gutiérrez-Ortiz et al (2014) [119]
CHP Electrical Efficiency	33	%	Clark (2013) [129]
CHP Heat Efficiency	43	%	Clark (2013) [129]
Net CHP Electricity	514.80	kWh/tonne feed	-
Net CHP Heat	649.82	kWh/tonne feed	-
Climate Impact			
Conversion Factor (Biogas)	0.00022	$kgCO_2$-eq/kWh	Hill et al (2021) - DBEIS [158]
Conversion Factor (Grid)	0.21233	$kgCO_2$-eq/kWh	Hill et al (2021) - DBEIS [158]
Climate Impact of Grid Electricity	119.75	$kgCO_2$-eq/tonne feed	-
Parasitic Load per tonne Feedstock	96.69	kWh/tonne feed	-
Climate Impact of Parasitic Load	20.05	$kgCO_2$-eq/tonne feed	-
Feed Transport Emissions	3.46	$kgCO_2$-eq/tonne feed	Hill et al (2021) – DBEIS [158]
Total Climate Impact (AD)	23.51	$kgCO_2$-eq/tonne feed	-

Table 12: GHG Inventory for Scenario 2: Gas-to-Grid

Data	Value	Unit	Source
Scenario 2 - Upgrading to Biomethane for Injection to Grid for Space Heating			
Upgrading Energy	0.25	kWh/m^3	Singh et al (2011) [111]
Upgrading Energy	49.44	kWh/tonne feed	Singh et al (2011) [111]
Climate Impact			
Conversion Factor (Biomethane)	0.00521	kgCO_2-eq/kg biomethane	Hill et al (2021) - DBEIS [158]
Conversion Factor (Natural Gas)	0.20297	kgCO_2-eq/kWh	Hill et al (2021) – DBEIS [158]
Climate Impact of Natural Gas	404.86	kgCO_2-eq/tonne feed	-
Parasitic Load per tonne Feedstock	137.31	kWh/tonne feed	-
Climate Impact of Parasitic Load	28.99	kgCO_2-eq/tonne feed	-
Feed Transport Emissions	3.46	kgCO_2-eq/tonne feed	Hill et al (2021) – DBEIS [158]
Total Climate Impact (AD)	32.45	kgCO_2-eq/tonne feed	-

Table 13: GHG Inventory for Scenario 3: HGV Fuel

Data	Value	Unit	Source
Scenario 3 - Upgrading to Biomethane and Compression for HGV Fuel			
Upgrading Energy	49.44	kWh/tonne feed	Singh et al (2011) [111]
Compression Energy	69.22	kWh/tonne feed	Singh et al (2011) [111]
Climate Impact			
Conversion Factor (Biomethane)	0.10625	$kgCO_2$-eq/MJ biomethane	Hill et al (2021) - DBEIS [158]
Conversion Factor (Diesel)	73.57	$kgCO_2$-eq/MJ diesel	Bates et al (2014) [154]
Climate Impact of Diesel	518.38	$kgCO_2$-eq/tonne feed	-
Parasitic Load per tonne Feedstock	202.32	kWh/tonne feed	-
Climate Impact of Parasitic Load	42.36	$kgCO_2$-eq/tonne feed	-
Feed Transport Emissions	3.46	$kgCO_2$-eq/tonne feed	Hill et al (2021) – DBEIS [158]
Total Climate Impact (AD)	42.36	$kgCO_2$-eq/tonne feed	-

4.2. Model Development

A sample calculation for estimation of GHG emissions for all three scenarios is included in the appendixes.

Pre-treatment

A two-stage pre-treatment system was utilised for this process. For the first stage, a macerator was chosen to reduce the particle size of the feed for easier pumping and to increase surface area for faster digestion inside the CSTR, with an electrical requirement of 7.5kW [63] to process the chosen feed rate of 20tpd. An eccentric screw pump, with an electrical requirement of 7.5kW [63], was then selected to transport feed from the macerator to the second stage of pre-treatment a pasteurisation tank where the feed was heated to 70°C to eliminate any pathogens, requiring 36.3kW of heat energy by calculation (see appendixes). A heat exchanger was then required to cool the feed from 70°C to the operating temperature of the digester (39°C).

Digester

The chosen digester for this process was an insulated CSTR operating under mesophilic conditions at 39°C [67, 109, 111]. It was assumed that the CSTR was perfectly insulated therefore heat losses are not considered. The volume of the reactor was assumed to be 660m^3 [67]. The CSTR was mixed through gas circulation at 5W/m^3 [157], requiring 3.3kW for a 660m^3 reactor. It was assumed the methane yield was 499.11m^3/tonne VS and the biogas composition was 65% methane [88].

H_2S Removal

Following digestion, trace amounts of H_2S must be removed before the biogas can be utilised due to corrosion. While it is not considered in the mass balance, the energy requirements for removal are considered in the energy balance. The chosen method was an activated carbon column which required 12kW [119].

CHP Unit

Inside the CHP unit biogas is combusted to provide heat and electrical energy. From literature, a 1000kW CHP unit has an electrical efficiency of 33%, a heat efficiency of 43% and losses of 24% [129]. 392.9kW of electricity was produced of which 45.3kW was diverted back into the process while the remaining 335.9kW was sold to the grid, whereas 512kW of heat was produced, of which 29.3kW was used to heat the pasteuriser and 29.9kW [67] was used to maintain the CSTR at 39°C. As there is no feasible way to store heat, any excess heat not required by the process went to waste.

Biogas Upgrading

For scenarios 2 and 3, the biogas must be upgraded to biomethane before it can be utilised. The chosen upgrading pathway was water scrubbing requiring 0.25 kWh/m^3 biogas [53] of electricity. Note that it was assumed that H_2S will also be removed during this process, therefore a separate unit for H_2S removal was not required. Following upgrading, biomethane was injected into the grid for scenario 2, or compressed and used as HGV fuel in scenario 3. For compression, it was assumed that 0.35kWh/m^3 biogas was required [111].

5. Results

An overview of the results of the LCA are outlined in the graphs below. Additional graphs and tabulated data are found in appendixes (Appendixes J and K).

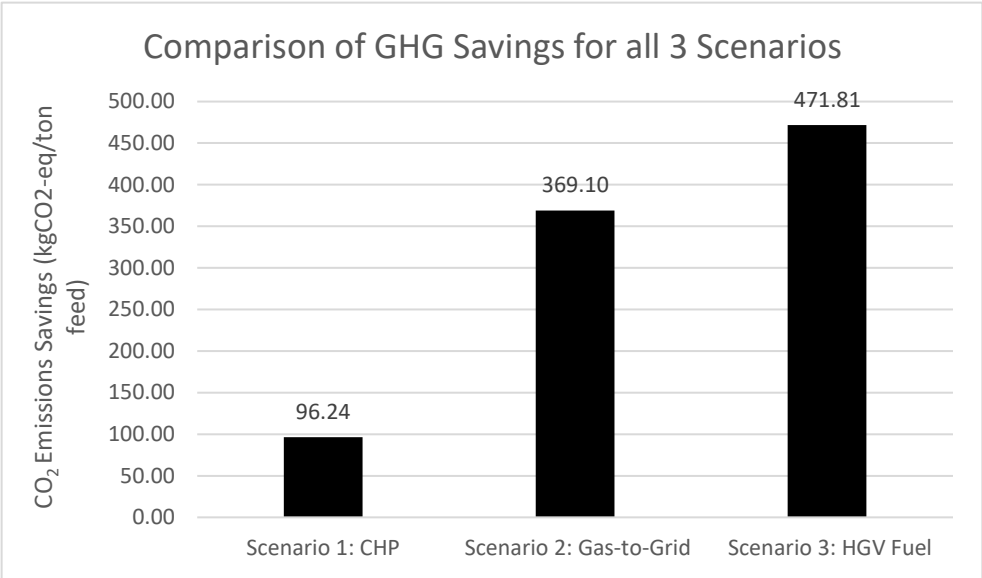

Figure 9: Comparison of GHG Emissions Savings for all 3 Utilisation Scenarios

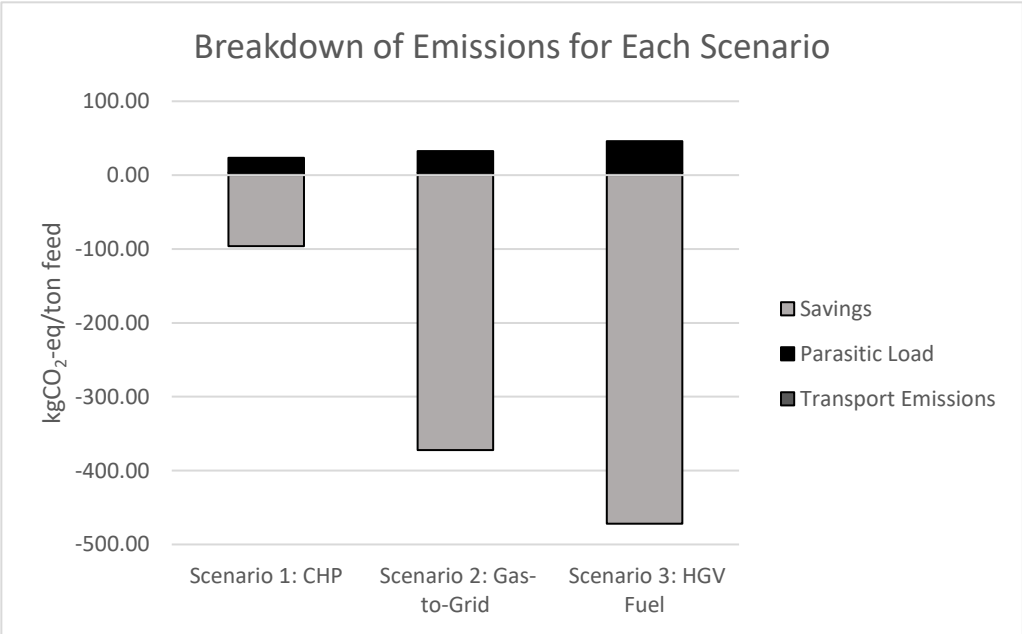

Figure 10: Breakdown of GHG Emissions for all 3 Utilisation Scenarios

From Figure 7. the largest reduction in GHG emissions is through Scenario 3 with savings of 471.81 $kgCO_2$-eq/tonne feedstock. Despite having a large additional energy requirement compared to Scenario 2 in the compression of biomethane to 200 bar, diesel is a far more polluting fuel than natural gas [158], accounting for the large difference in GHG savings. While Scenario 2 also offers substantial reductions of 369.10 $kgCO_2$-eq/tonne feedstock, there are limitations to be considered. Regulation and infrastructure for injection biomethane to the grid in NI is lacking, with the natural gas grid providing poor coverage to rural areas where most AD facilities are located [130]. Scenario 1 offers the lowest GHG savings of 96.24 $kgCO_2$-eq/tonne feedstock, owing to the gradual decarbonisation of the UK grid. In 2015 the UK grid CO_2 conversion factor was 0.46219$kgCO_2$-eq/kWh [159] whereas in 2021 this had fallen by more than half to 0.21233$kgCO_2$-eq/kWh [158]. With further efforts to decarbonise electricity generation in the future, it can be expected that the GHG savings through this utilisation pathway will diminish further over time.

Figure 8. provides a breakdown of the sources of GHG emissions for all three pathways, with the parasitic load of the plant as the largest contributor regardless of pathway. The energy requirement for Scenario 2 and Scenario 3 is greater than that of Scenario 1 due to the increased energy requirements for biogas upgrading and compression.

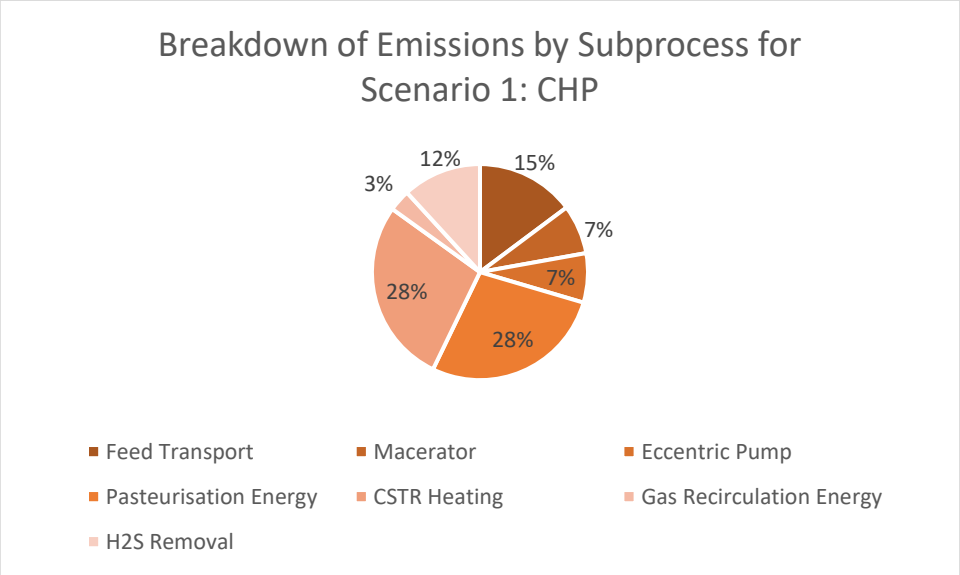

Figure 11: Breakdown of GHG Emissions for Each Subprocess for Scenario 1

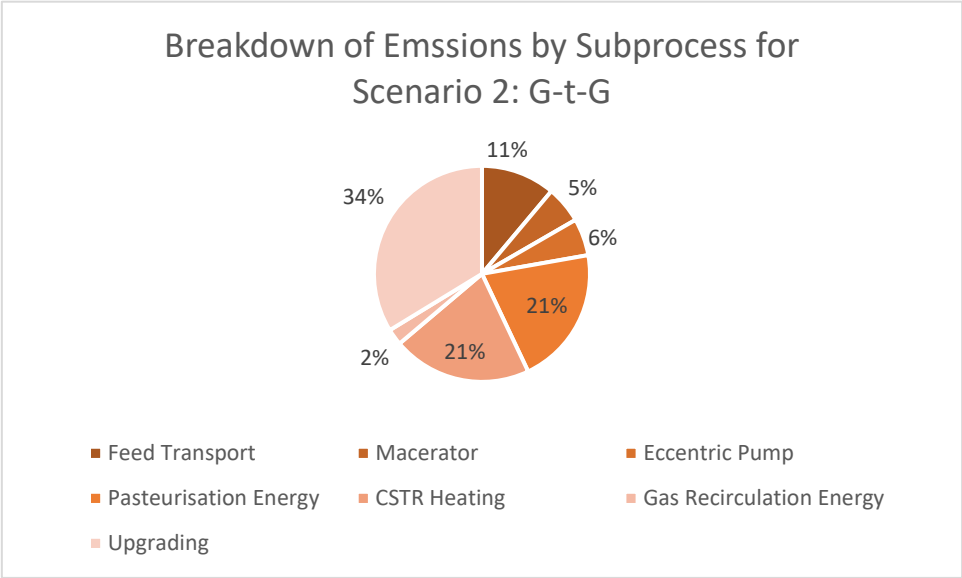

Figure 12: Breakdown of GHG Emissions for Each Subprocess for Scenario 2

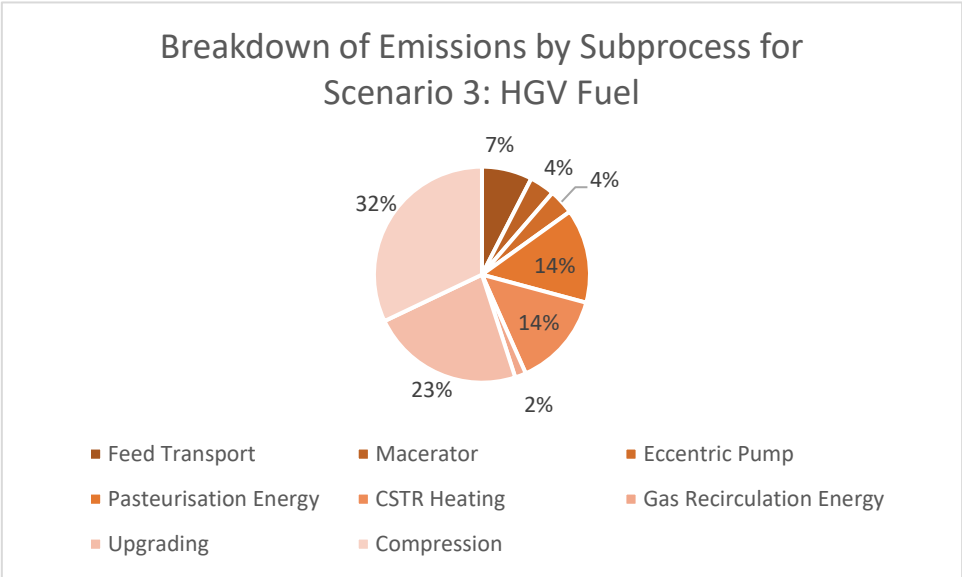

Figure 13: Breakdown of GHG Emissions for Each Subprocess for Scenario 3

Figures 9-11. provide a breakdown of the contributions each subprocess makes towards overall GHG emissions of the process. While CSTR heating and pasteurisation contribute most for Scenario 1, the electrical requirements for biogas upgrading and compression in Scenarios 2 and 3 contribute significantly more CO_2 emissions overall. Additional graphical illustrations and tabulated data for these breakdowns are provided in the appendixes (Appendix L-O).

5.1. Sensitivity Analysis

Varying Feedstocks

The chosen feedstock used for anaerobic digestion has a large impact on the overall efficiency of the process. Different feedstocks have different compositions of carbohydrates, lipids and proteins which affects the methane yield [160]. For this sensitivity analysis, two different feedstocks were considered in addition to the chosen feedstock of poultry waste. The first option was household food waste, with a methane yield of 346.20m^3/tonne VS [161]. Cattle SHW with a methane yield of 641.60m^3/tonne VS [63] was the second option selected for sensitivity analysis, comparing against the chosen poultry SHW with a methane yield of 499.11m^3/tonne VS [88]. Below are graphs outlining the effect different feedstocks have on energy generated and overall process emissions. Additional graphs and tabulated data for illustration are included in the appendixes (Appendixes P).

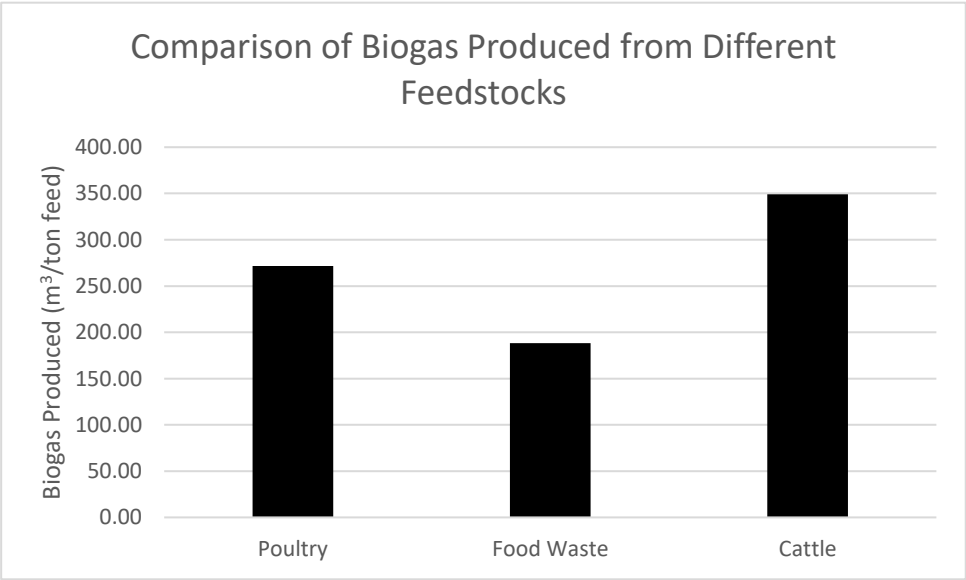

Figure 14: Effect of Feedstock on Biogas Production

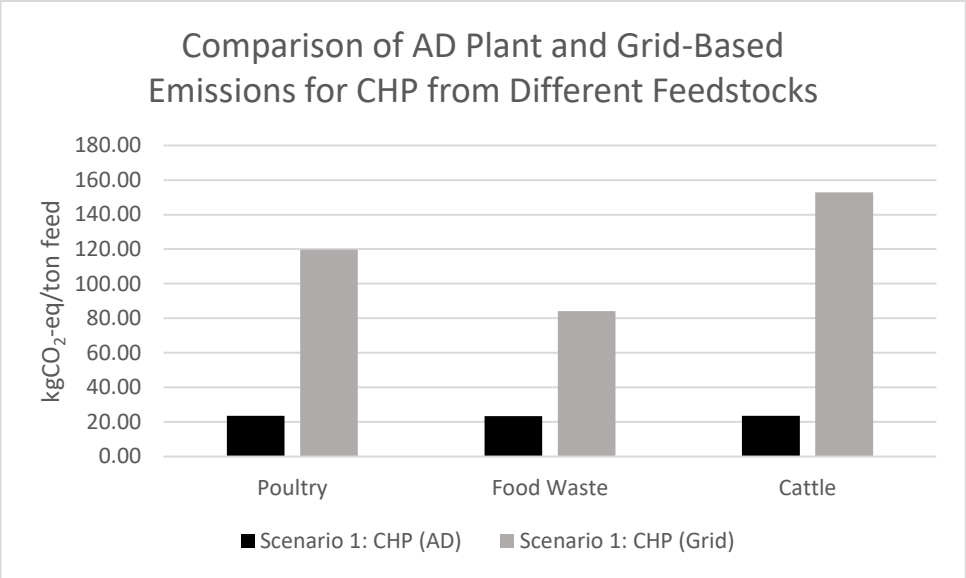

Figure 15: Effect of Feedstock on GHG Emissions

As expected, the lower methane yields returned worse savings on GHG emissions and less energy generated overall. Of the three pathways, Scenario 1 is the most sensitive to a change in feedstock and therefore a change in methane yield. This is because the energy requirements of Scenario 2 and Scenario 3 are dependent on the amount of biomethane generated, therefore increasing the yield also increases the energy requirement. As there is less than 1% in difference of CO_2 savings between the chosen feedstock and the higher yield feedstock, there is not significant enough of an improvement to justify choosing cattle SHW over poultry waste, due to the increased energy requirements outlined above. Furthermore, this analysis assumed the same distance travelled from waste processing plants to the AD plant. As mentioned earlier in this report, there were not any cattle SHW sources locally with a large enough output to satisfy plant demand. Increased transportation emissions would further offset the increased biogas yield from cattle feedstock.

The second analysis considered was the effect of changing the biogas upgrading method. The following range of technologies was considered: PSA, water scrubbing, physical scrubbing, chemical absorption, and membrane separation [53]. An average value from each range displayed in [53] was calculated for use in this analysis. A graph outlining the effect of upgrading method on GHG emissions is included below. Additional tabulated data is included in the appendixes (Appendix Q).

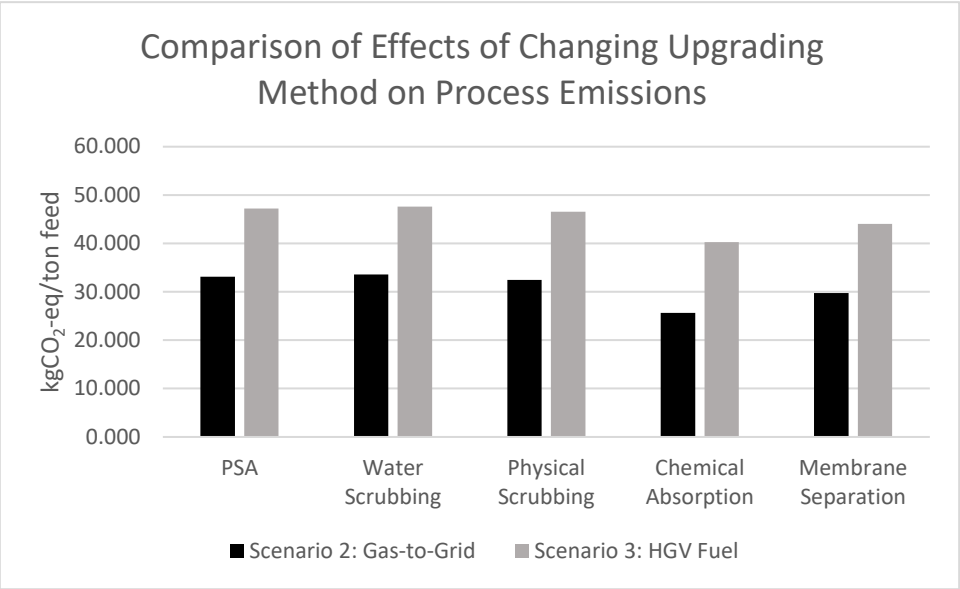

Figure 16: Effect of Upgrading Method on GHG Emissions

From the results of the analysis, it is seen that the chosen upgrading method has a low overall effect on the total CO_2 savings for the process compared to the grid. While the electricity requirements vary from 0.1 to 0.275 kWh/m^3, the largest range in GHG savings is 25.619 to 33.591 kgCO_2-eq/tonne feed for Scenario 2 and 40.272 to 47.621 kgCO_2-eq/tonne feed for Scenario 3. This is congruent with previous research [64], suggesting that the electricity requirements for biogas upgrading is only a small contributor to overall plant emissions and that the method chosen does not significantly matter with regards to overall emissions.

6. Discussion

This study aimed to provide both a cradle-to-grave assessment of the GHG emission reduction potential of 3 biogas utilisation pathways, as well as the substitution of SHW for traditional AD feedstocks within the context of the NI regional biorefinery concept. This was achieved through the pursuit of the fundamental objectives of quantifying regional SHW feedstocks and their workable aptitude for AD, the analysis of GHG emission reduction potentials under varying unit configurations based on an operational plant baseline scenario to provide operational recommendations for new market entrants, and the comparison of the optimal conditions as found in the study with current-state infrastructure to inform NI bioeconomic policy.

Utilising a modified process configuration of the AFBI Hillsborough plant as a baseline scenario [67], it was found that the substitution of SHW produced demonstratable GHG emissions savings under each scenario in comparison with the slurry and grass silage feedstock currently in use operationally (due in part to the high methene yield per tonne of SHW organic content, and the local availability of poultry SHW from theoretical supplier Moy Park, Craigavon). Furthermore, each scenario demonstrated GHG emissions savings in comparison with electricity from the grid apart from baseline scenario. Interestingly, this is in-part not congruent with the most pertinent study to date in this field, [64], which evaluated the same utilisation pathways for the same operational AFBI plant (although it did not modify the unit configuration or feedstock for its baseline scenario). The study found that the operational AFBI plant (utilising cattle slurry and silage as feedstock) demonstrated higher GHG emission reduction savings than utilising electricity from the grid. Notably, this illustrates the significance of the definition of a representative functional unit within life cycle analysis: in [64], the unit of calculation was defined as 1 MWh of electricity injected into the grid, whereas this study assessed scenario viability in terms of kWh/tonne of feedstock, materially altering the conclusions reached through a life cycle analysis as stipulated in [81, p. 20]. While the rationale for this choice as asserted in [64, p. 8] is the focus on an evaluation of the product of each scenario as an energy carrier, this study's conclusions are better aligned to its stated objectives of analysing regional feedstock substitution feasibility for the development of holistic policy recommendations. Materially, the difference in this conclusion can be explained by the low biogas yield achieved by utilising slurry and silage (20m^3/m^3 of slurry or ~19 $m^{3/}$/tonne of feedstock) [69] compared with 271 m^3/tonne of feedstock demonstrated in this study (owing to the high lipid content as evaluated in section 3.2.1) and the higher overall TS content (8.9 tpd for the SHW model as opposed to 1.42 tpd for the AFBI feedstock (for the same feed capacity). When considering the policy drivers for the valorisation of waste, the practical considerations of handling waste material in an operational setting, and the emissions burden of transporting the feedstock to the refinery, optimising biogas production per unit mass of feedstock is the best practicable methodology for analysing biorefinery systems.

A sensitivity analysis on the optimisation of FW feedstock selection demonstrated that while utilisation of poultry SHW resulted in a higher GHG emission reduction potential than household FW, cattle SHW showed greater potential for methane yield per tonne of feedstock and resultant GHG emission savings. Not only is this consistent with findings in [63] but the

utilisation of cattle SHW as a feedstock would contribute significantly to reducing the GHG emissions of the agri-food sector as it relates to the political drivers for this study (stated in section 1): Ireland is the biggest exporter of beef in the EU and the 5th largest in the world [63, p. 1]. However, as illustrated in section 3.2.1, quantifying the mass of SHW produced by NI abattoirs (as well as their stream characterisations) presented a barrier to the utilisation of this feedstock in the model, and will materially inhibit its uptake in a workable scenario, necessitating the development of transparent industrial publications of effluent streams from all FW producers through policy intervention. However, operational data from a Swedish plant processing SHW resulted in the outlining of the following limitations: increased ammonia production (and methanogenic inhibition) due to the high protein content of SHW, and high pump work-rate due to decrease flow resulting from high TS content. Its recommendations for proper management of the AD of SHW include the following:

- 50% (w/w) co-digestion with cattle manure
- pre-treating the waste with formic acid to reduce the odour impact on workers
- the addition of ferrous chloride and hydrochloric acid for the reduction of H_2S accumulation pH management
- the addition of an additive (KMBI) to increase stability within the digester
- thermal pre-treatment before pasteurisation to reduce clogging and ease the pumping load
- longer HRT (45-55 days to enable the establishment of mesophilic syntrophic acetate oxidizing culture
- bi-phase digestion for increased methane yield (congruent with [162] [163] [164])

Each of these mitigation systems would increase the process intensity of utilising SHW as a feedstock, requiring the further study of an operation pilot plant to iteratively engage with primary data in validating the viability of SHW within the NI biorefinery concept.

The regional effects on SHW utilisation in AD are exemplified by the results of this LCA, identifying the GHG emissions from the transportation of feedstock to the refinery as a major contributor to the environmental performance of the plant. Moy Park, Craigavon is only 32.1km from the AFBI site, but this study found that the transportation burden from this site contributed to 14.8%, 11.1% and 7.6% of GHG emissions (scenarios 1, 2 and 3 respectively).

[64] asserts that upon consultation with industrial partners, it was found that feedstock selection was largely driven by contractual factors, leading to transportation distances

exceeding 200km in practise. An option for the further development of the NI biorefinery concept is the advancement of local infrastructural and economic networks, better connecting waste producers with waste-to-energy facilities, as well as research into the effects of utilising biodiesel (as produced in scenario 3) in the transportation of feedstock to the plant itself (in accordance with circular economy theory [37]. Further contractual considerations include agreed pre and post-processing obligations as regards to the system boundaries of the AD refinery. In [85], the SHW is macerated and chemical treated at the SH site to facilitate handling of the waste at the refinery, whereas the LCA model assumed in-situ pre and post-processing for all sub-processes. Specifically, the treatment of SHW necessitating pasteurisation places an extra burden on new entrants, further requiring the evaluation on an operational system.

These considerations indicate the need for the assessment of the economic viability of the development of such a system for new entrants, which was not considered in this study. Although the low cost of feedstock (in the utilisation of waste) and revenue generated from selling commodified biogas and digestate may offset some of the economic burden of the process intensity described herein, the assumption of steady-state conditions discounts the cost burden of plant commissioning and start-up. Additionally, the energy required to raise the feed from ambient conditions to pasteurisation is dependent on seasonal variation [67], which was not considered in the LCA model (15°C temperature assumed year around). A sensitivity analysis could be performed to assess the effect of seasonal temperature change on the energy requirement, and the development of an end-to-end economic feasibility model should be developed in tandem with the ongoing research effort into the development of the regional NI biorefinery concept.

A well-documented limitation of operational AD refineries is methane leakage (fugitive emissions). From a survey of 10 biogas plants, an average methane leakage rate of 15.9kg/hr was quantified [56]. With a GWP potential over 100 years of 28-36 times greater than that of CO_2 [165], this is a significant source of emissions not considered in this LCA, warranting further operational study.

Further limitations of this study include the variability incumbent in the calculation of GHG emission reductions. In this study, emissions were quantified through the UK government database of GHG conversion factors [158]. These factors are based on the global warming potential (GWP) of individual GHGs, then converted to a CO_2-equivalent factor. Continued

evaluation of GWP as a metric of emissions has found uncertainties in the figures, with revisions being made as a better understanding of the long-term responses in the carbon cycle is developed [166] ([64] discusses the effects of the conversion factor selection on LCA conclusions).

In evaluating the performance of each scenario in accordance with this study's objectives, it was found that scenario 3 (upgrading biogas to biodiesel (LNG) resulted in the largest GHG emissions savings per tonne of feedstock (471 $kgCO_2$ compared to 96 $kgCO_2$ and 369 $kgCO_2$ respectively for scenarios 1and 3: the contributing factor being identified as the high emissions burden of diesel utilisation by HGVs (73 $kgCO_{2\ eq}$/MJ for diesel compared with 0,1 $kgCO_{2\ eq/MJ}$ for biodiesel), further solidifying the recommendation outlined above that research be undertaken to evaluate the impact of substituting feedstock transportation diesel for biodiesel in this LCA. This being despite scenario 3 requiring the largest process intensity due to the compression load (requiring 14.7 $kgCO_2$/tonne of feed.) There is currently little infrastructure available to facilitate expansive uptake of this utilisation pathway: there are only 7 LNG stations operational in Ireland, although 7 public stations have been commissioned. As LNG stations are privately owned and scarce, there is limited availability of biomethane as an alternative fuel [152], warranting further research into the expansion of required infrastructure.

Finally, the limited scope of this study presents the largest potential risk for the validity of its conclusions. Only 1 impact category (climate change) was assessed in this study through the lens of calculating the GHG emission reduction potential of the system. However, [64] was able to utilise Midpoint Hierarchical V1.13 software to implement the ReCiPe method in evaluating a further 17 impact categories: ozone depletion, terrestrial acidification, fresh water eutrophication, marine eutrophication, human toxicity, photochemical oxidant formation, particulate matter formation, terrestrial ecotoxicity, freshwater ecotoxicity, marine ecotoxicity, ionising radiation, agricultural land occupation, urban land occupation, natural land transformation, water depletion, metal depletion and fossil depletion. This software was not available for use in this study, and the inclusion of these considerations (together with an economic evaluation) would give rise to a more representative, workable recommendation to industrial partners and policy members. A further impactful consideration as regards to the scope of this study is related to the environmental impact analysis of the production of the selected feedstock. Although valorisation of SHW presents a viable operational endeavour under current waste production rates, an evaluation commissioned by the NI Minister for

Agriculture found that reaching NI's GHG emissions targets would require an 86% reduction in ruminants, significantly impacting the agri-food sector in the region, equating to a 98% reduction in sheep and beef farming [70]. This is owing to methane's association with ruminant livestock production accounting for two-thirds (68%) of Irish agricultural GHG emissions [167], indicating perhaps the most significant area for related further research.

7. Conclusions

This study concludes that the pursuit of investment into the utilisation of biogas from SHW as biodiesel for HGVs is the optimum utilisation pathway for biogas and can provide a significant offset to both the GHG emissions, and the waste-management burden created by the agri-food sector, while also providing significant emissions savings when compared to electricity for the grid. SHW is proven to be a viable, high-value feedstock option for the AD process, but process limitations and infrastructural/political deficiency require further study in an operational setting.

Recommendations for further study

- The valuation of life-time emissions of feedstock production and its effects on GHG emission reduction potential
- The undertaking of an iterative, operational approach with continually refined models in an operational plant
- Widening the scope of study to consider further impact categories
- Research into frameworks for the development of infrastructure for biodiesel utilisation as fuel for HGVs in NI
- The furthering of efficiency gains in biomethane compression technologies to lower the GHG emissions burden of this sub-process

7.1. *(Non-exhaustive) List of Assumptions Summary*

- The plant was assumed to be operating under steady-state conditions, therefore emissions from planning, construction, materials, decommissioning, and land reclamation were not considered.
- Assumed 330 days of operation per year to allow for maintenance and inspection.
- A feed rate of 20tpd was assumed based on the AFBI Hillsborough technical report [67].
- As the feed had TS of 44.9% [88] it was assumed the remaining 55.1% was water, through which the heat capacity of the feed was calculated.
- The average ambient temperature for calculation of heat duties was assumed to be 15°C.
- Inside the digester, it was assumed all volatile solids were converted to biogas and all remaining mass was converted to digestate.
- While H_2S is present in biogas at concentrations ranging from 1000-3000ppm [168], the amount present was considered negligible when applying a mass balance over the AD plant.
- The biogas produced was assumed to contain only CH_4 and CO_2 for all mass balance calculations.
- All the CH_4 contained in the biogas was assumed to undergo complete combustion inside the CHP unit, with 100% stoichiometric conversion of CH_4 and O_2 to CO_2 and H_2O.
- All sub-processes, unless otherwise stated such as the CHP unit, were assumed to be lossless and operating at 100% efficiency.
- The calorific value of biogas was assumed to be 22MJ/m^3 [169].
- The calorific value of biomethane was assumed to be 36MJ/m^3 [148].
- All H_2S and other trace compounds were assumed to be removed during the upgrading process.

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Tables and Figures (Sequential)

Table1: Statement of Individual Contributions

Name	Student Number	Contributions
Anastasia Perysinakis	401778877	Abstract, 1, 2, 3.1.1, 3.1.4, 3.2, 4.1, 5, 6, 7
Hanna Blomkvist Lager	40184784	3.1.1, 3.1.2, 3.1.3, 3.2, 4.1, 6
Sam Balmer	40176908	3.1.1, 3.1.2, 3.2, 4.1, 4.2, 5, 6
Michael Gamble	40225575	
Clyde Scott	40234545	(3.1.4)
Connaire Henderson	402	(3.1.1, 3.1.2)
Kannappan	402	(1, 3.1.4)

N.B. (1) Contributions in parenthesis denote work under duplicate titles submitted separately to this study that is not featured in this paper.

N.B. (2) Shaded line-items denote previous contributing members who contributed intangibly to the undertaking of this study, but without notable significance to the works herein.

Table 14: Examples of raw biogas compositions

	Source[a]			
	Kapoor et al (2020)[47]	Benato et al (2019)[48]	Kuo et al (2017)[49]	Muñoz et al (2015)[50]
	Primary Feedstock[b]			
	(Average)	Cattle sewage	Commercial Food Waste	Landfill
Constituent	**Concentration (v/v)**			
Methane	40-60%	53 ±8%	64.6%	35-65%
Carbon Dioxide	35-55%	44 ±7%	34.9%	15-50%
Moisture	1-5%	<2%	-	<5%
Nitrogen	0-5%	1.9±0.3%	0.7%	5-40%
Hydrogen	Trace	<0.5%	<1.3%	<3%
Hydrogen Sulphide	0.1-3%	645ppm	163ppm	<100ppm
Oxygen	<2%	0.4±0.06%	<0.1%	<5%
Trace gases	<2%	<2%	<2%	<250ppm
Ammonia	0-500ppm	15±4.7ppm	Trace	<5ppm

[a] Sources sampled from both industrial examples and simulation-based approaches

[b] Co-digestion employed in all cases where a feedstock is identified, the most significant of which (according to the authors) is displayed above

[c] Calculated by conversion for consistency

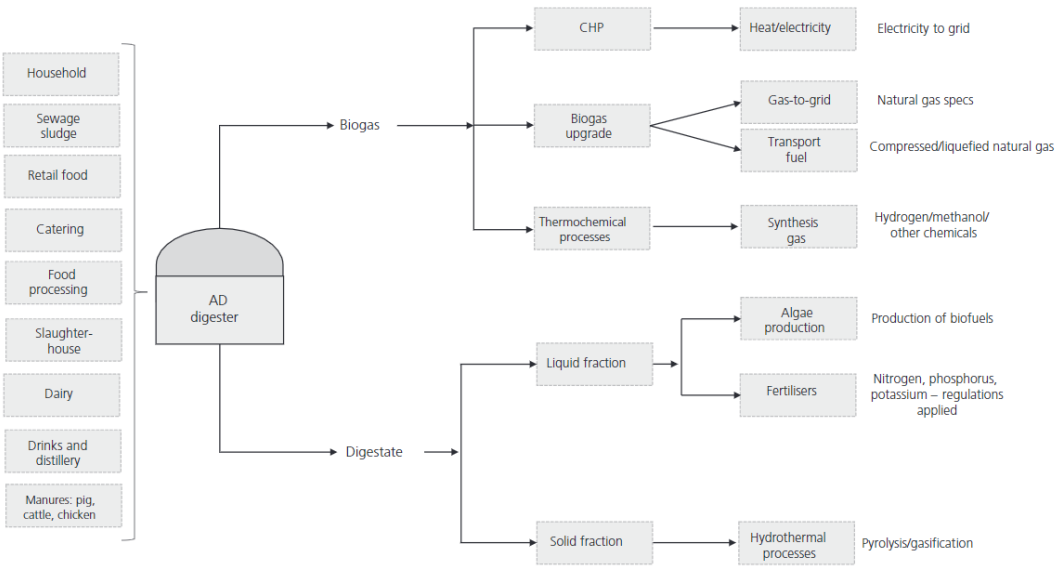


Figure 17: Feedstocks and utilisation pathways for AD products in a large-scale biorefinery [42]

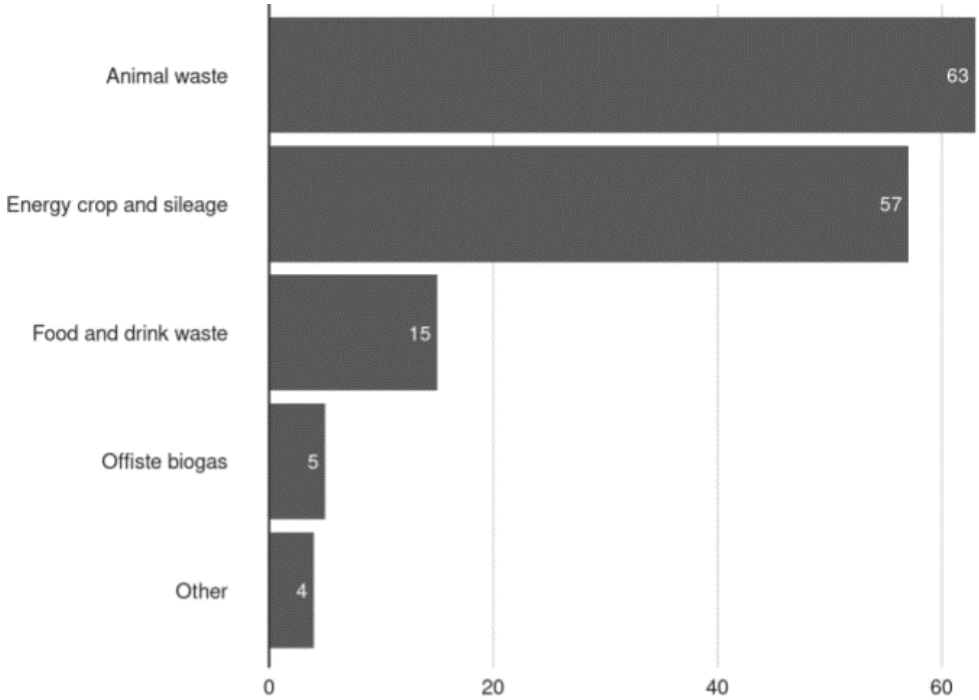


Figure 18: AD feedstock usage in NI.

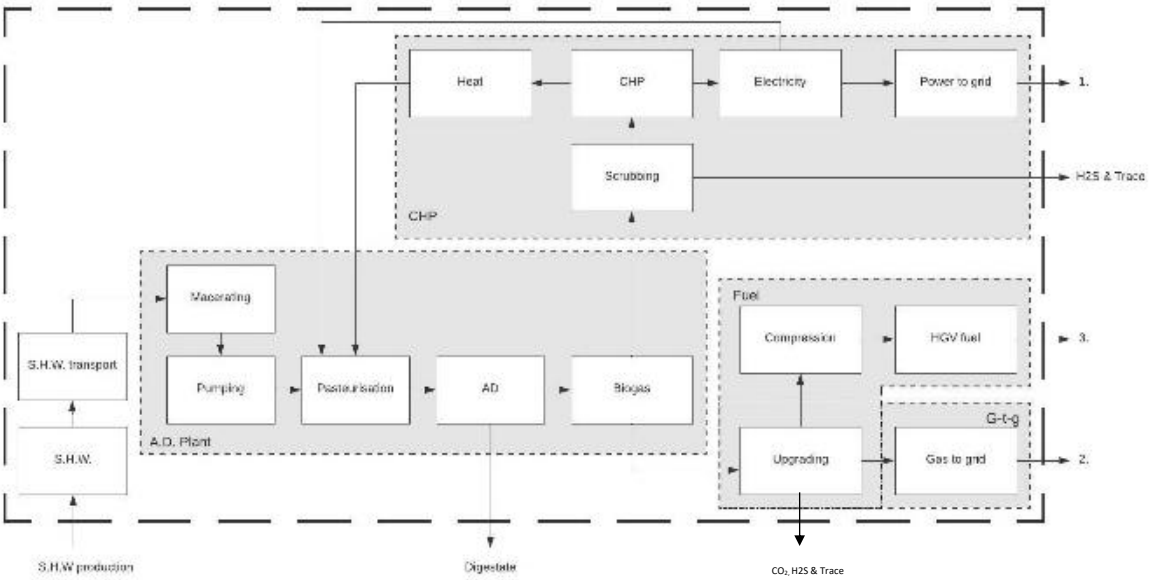


Figure 19: Overall system boundary including all 3 scenarios under investigation. Shaded sections denote an internal boundary for each sub-system under evaluation: the AD plant and systems supporting each of the 3 scenarios ((CHP (1), G-t-g (2), and HGV fuel (3)).

Table 15: Summary table delineating inputs and outputs of each unit illustrated in the system PFD

Process Unit	Inputs	Unit	Outputs	Unit
		AD Plant		
Macerator	SHW	tonne	SHW (macerated)	tonne
Pasteuriser	SHW (macerated)	tonne	SHW (pasteurised)	tonne
	Heat (parasitic load)	kWh/tonne feed		
	Power (parasitic load)	kWh/tonne feed		
AD Unit	SHW (pasteurised)	tonne	Biogas (raw)	m^3/tonne feed
			Digestate	tonne
		Scenario 1		
Scrubbing Unit	Biogas (raw)	m^3/tonne feed	Biogas (clean)	m^3/tonne feed
			H_2S	m^3/tonne feed
CHP	Biogas (clean)	m^3/tonne feed	Heat	kWh/tonne feed
			Power	kWh/tonne feed
		Scenario 2		
Upgrading Unit	Biogas (raw)	m^3/tonne feed	Biomethane	m^3/tonne feed
		Scenario 3		
Compressor	Biomethane	m^3/tonne feed	Biodiesel (LNG)	m^3/tonne feed

Table 16: Characterisation of sampled slaughterhouse waste flow streams [84]

Solid waste and solid mixtures (SW)	Cattle meat and fat	Pig meat and fat	Confiscates (lungs, livers and kidneys)	Pig stomach	Rumen content	SW1	SW2
TS (g kg^{-1})	885.58 ± 4.81	564.54 ± 19.18	244.89 ± 2.89	182.89 ± 7.88	116.86 ± 5.04	527.13 ± 7.47	374.53 ± 7.50
VS (g kg^{-1})	854.14 ± 4.80	557.33 ± 20.79	219.67 ± 3.57	179.72 ± 7.88	108.68 ± 4.35	521.52 ± 7.70	366.94 ± 7.23
TKN (g kg^{-1})	3.21 ± 0.54	13.80 ± 0.36	26.32 ± 10.30	12.38 ± 0.69	1.32 ± 0.18	10.75 ± 0.11	11.69 ± 0.11
NH_4^+–N (mg kg^{-1})	148.51 ± 2.06	389.70 ± 64.67	1512.37 ± 19.53	1645.81 ± 15.00	60.83 ± 2.05	274.51 ± 77.64	537.55 ± 195.10
Protein$_{estimated}$ (g kg^{-1})	*19.14*	*83.80*	*155.04*	*67.10*	*7.88*	*65.48*	*69.73*
Fat$_{soxtec}$ (g kg^{-1})	762.79 ± 2.39	467.30 ± 14.90	46.60 ± 1.50	86.64 ± 0.01	18.42 ± 1.18	432.07 ± 4.80	249.38 ± 2.45
COD (g kg^{-1})	n.d	n.d	n.d	n.d	n.d	1356.03 ± 21.66	849.24 ± 77.99
COD$_{estimated}$ (g kg^{-1})	*2294.47*	*1473.46*	*385.40*	*376.85*	*152.01*	*1363.66*	*871.01*
Liquid waste and final mixtures (M)	**Waste blood**	**Screening waste**	**DAF sludge**		**Wastewater**	**M1**	**M2**
TS (g kg^{-1})	196.60 ± 0.01	242.30 ± 2.72	95.65 ± 0.78		0.93 ± 0.02	79.64	36.00
VS (g kg^{-1})	184.04 ± 0.03	233.34 ± 2.61	82.66 ± 0.76		0.53 ± 0.03	77.94	34.36
TKN (g kg^{-1})	31.70 ± 3.01	12.40 ± 0.32	5.87 ± 1.52		0.15 ± 0.01	1.89	1.67
NH_4^+–N (mg kg^{-1})	133.00 ± 9.90	1502.21 ± 20.14	861.00 ± 9.90		84.00 ± 2.30	149.09	135.17
Protein$_{estimated}$ (g kg^{-1})	*197.34*	*68.12*	*31.29*		*0.41*	10.88	9.59
Fat$_{soxtec}$ (g kg^{-1})	n.d	107.11 ± 0.64	50.30 ± 0.05		0.10 ± 0.01	63.29	20.92
COD$_{estimated}$ (g kg^{-1})	*262.73*	*471.18*	*192.44*		*0.92*	*201.96*	*78.51*

Table 17: Livestock numbers by type (UK) [59]

	England	**Scotland**	**Wales**	**NI**
Total livestock numbers	Cattle – 5,146,000 Sheep- 15,390,000 Pigs – 3,776,000 Poultry – 138,850,000	Cattle – 1,730,000 Sheep – 6,670,000 Pigs – 319,300 Poultry - 14,900,000	Cattle – 1,137, 399 Sheep – 10,037,474 Pigs – 24,503 Poultry - 7,515,804	Cattle – 1,611,776 Sheep – 1,986,932 Pigs – 674,428 Poultry – 24,780,381
Approximate crude livestock stocking densities per hectare (livestock nos ÷ by total farmed area)	Cattle – 0.56 Sheep – 1.6 Pigs – 0.41 Poultry – 15.15	Cattle – 0.30 Sheep – 1.17 Pigs – 0.05 Poultry – 2.63	Cattle – 0.71 Sheep – 6.29 Pigs – 0.01 Poultry – 4.7	Cattle – 1.57 Sheep – 1.94 Pigs – 0.65 Poultry – 24.21

Table 18: Biodegradation of different animal wastes in anaerobic mono-digestion [86]

Substrates	T(°C)	Operation Mode and conditions	Methane yield (Y_{CH_4})	Degradation efficiency (%)
Slaughterhouse waste				
Solid cattle	55	Batch (2L)	582-681 (mL gVS^{-1})	n.a
	55	CSTR (3L), HRT= 25 days, OLR=0.9 gVS $L^{-1}d^{-1}$	410 (mL gVS^{-1})	n.a
Solid cattle and pig	35	CSTR (2L), HRT= 30 days	60 (mL gVS^{-1})	34.5 (VS red)
Solid and liquid pig/cattle mixtures	35	Batch (1L)	274-302 (mL $gCOD^{-1}$)	86-92 (COD red)
Iberian pig	38	CSRT (2L), HRT=23.5 days	17.84 (m^3 m^{-3}_{Feed})	78.59 (COD red)
Pig	55	Batch (0.5 and 2L)	230-620 (mL gVS^{-1})	50-100 ($Y_{CH_4}/Y_{CH_{4Theoretical}}$)
	35	Batch (1.2L)	460 (mL gVS^{-1})	76.6 (COD red)
Poultry	35	Batch (1.2L)	580 (mL gVS^{-1})	55.2 (COD red)
	35	CSTR (2L), HRT= 50 days, OLR=0.8 gVS $L^{-1}d^{-1}$	520-550 (mL gVS^{-1})	n.a
	34	CSTR (3L), HRT= 50, 36, 25 days, OLR=0.9, 1.16, 1.70 gVS $L^{-1}d^{-1}$	600-700 (mL gVS^{-1})	77-79 (VS red)

Table 19: Characteristics of poultry waste streams. [88]

	DAF	SO	PSO	SHWM
TS[a]	11.7 (0.59)	40.3 (3.69)	44.9 (2.17)	24.2 (0.21)
VS[a]	93.4 (0.02)	96.6 (0.70)	96.6 (0.70)	94.8 (0.11)
Fat[b]	2.7	36.1	36.1	24.3
Protein[b]	53.5	34.0	34.0	40.9
Carb[b]	37.2	26.5	26.5	29.6
C[b]	51.6	55.3	55.3	54.0
H[b]	7.8	9.4	9.4	8.9
O[b]	25.2	26.7	26.7	25.5
N[b]	8.8	5.1	5.1	6.4

a % (standard deviation).
b % of TS.

Table 20: Results of BMP assays [88]

Substrate	IP[a] (days)	V_0 ($mlCH_4$ gVS^{-1})	$V_{0d\ max}$[b] ($mlCH_4$ gVS^{-1} d^{-1})	80% of V_0 (days)
DAF	30	261.35 ± 20.39	46.19 ± 1.34 (1)	10
SO	30	499.11 ± 35.44	32.80 ± 0.39 (17)	21
PSO	50	501.13 ± 40.59	45.81 ± 0.14 (1)	36
SHWM	50	594.59 ± 38.05	45.10 ± 1.42 (23)	32

Note: relative standard deviation applied where necessary.
[a] IP- Incubation period of BMP.
[b] ( ) day on which maximum daily V_0 occurred.

Table 21: Distances of Slaughterhouse Waste Sources from AFBI Hillsborough Site

Feedstock Source Site	Distance from AD Facility (AFBI Hillsborough Plant)/km	Feedstock Type
Moy Park, Dungannon	52.6	Poultry Waste
Moy Park, Ballymena	55.4	Poultry Waste
Moy Park, Craigavon	27.7	Poultry Waste

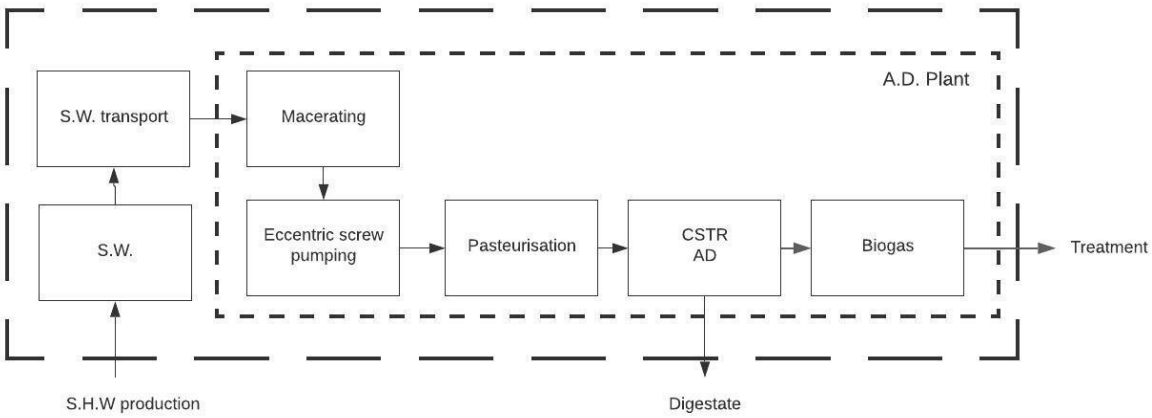


Figure 20: System boundary for the A.D. plant

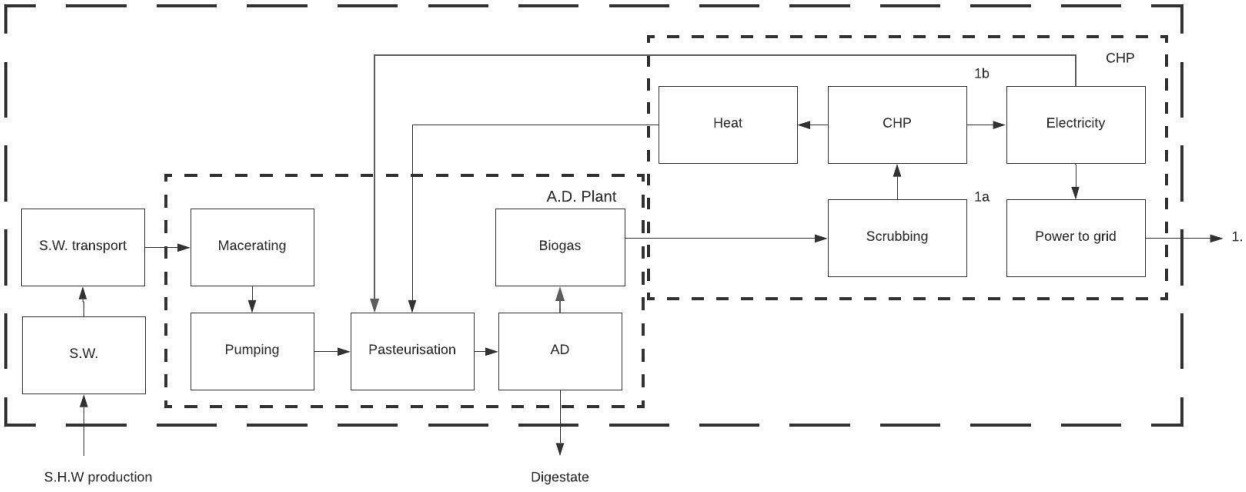


Figure 21: System boundary for scenario 1 (CHP, power to grid)

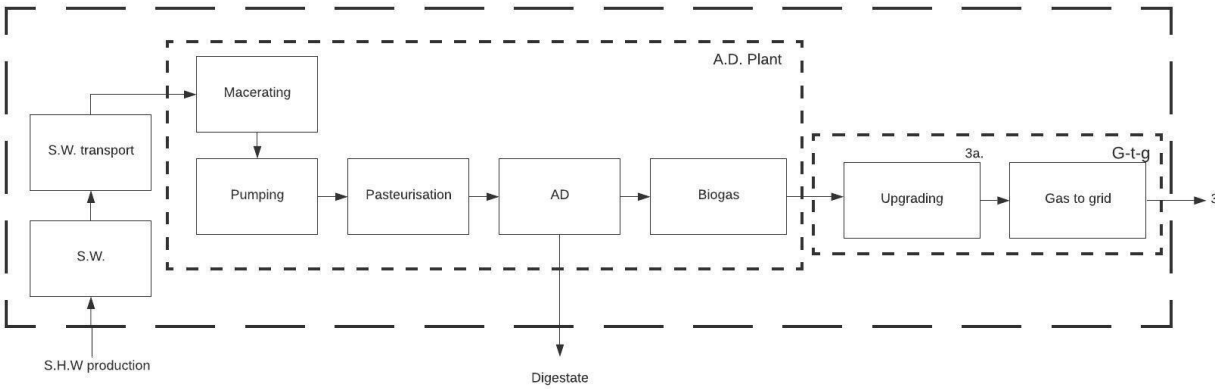


Figure 22: System boundary for scenario 2 (G-t-g)

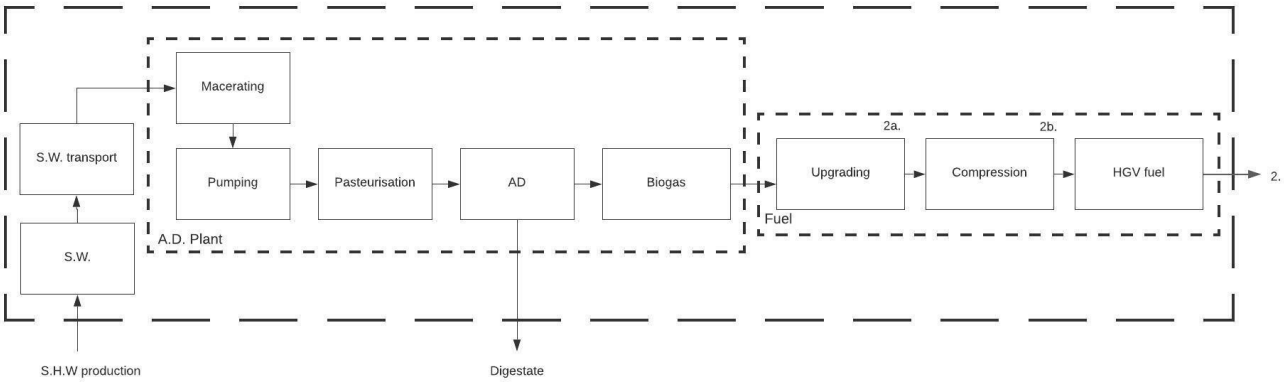


Figure 23: System boundary for scenario 3 (HGV Fuel)

Table 22: Input data for general AD Plant

Data	Value	Unit	Source
Input to AD Plant			
SHW	7300	tonne per year	Calculated
Macerator	8.14	kWh/tonne feed	Ware and Power (2016) [63]
Pasteurisation Energy	31.81	kWh/tonne feed	Calculated
CSTR Heating	32.00	kWh/tonne feed	Frost et al (2011) - AFBI Plant [67]
Gas Recirculation Energy	3.58	kWh/tonne feed	Mahmoodi et al (2020) [157]
SHW Methane Yield	499.11	m^3/tonne VS	Ware and Power (2016) [63]
Output of AD Plant			
Biogas	271.57	m^3/tonne feed	Calculated
Digestate	0.63310	tonne/tonne feed	Calculated

Table 23: GHG Inventory for Scenario 1: CHP

Data	Value	Unit	Source
Scenario 1 - Electricity to Grid from CHP Unit			
H_2S Removal Energy	13.02	kWh/tonne feed	Gutiérrez-Ortiz et al (2014) [119]
CHP Electrical Efficiency	33	%	Clark (2013) [129]
CHP Heat Efficiency	43	%	Clark (2013) [129]
Net CHP Electricity	514.80	kWh/tonne feed	-
Net CHP Heat	649.82	kWh/tonne feed	-
Climate Impact			
Conversion Factor (Biogas)	0.00022	$kgCO_2$-eq/kWh	Hill et al (2021) - DBEIS [158]
Conversion Factor (Grid)	0.21233	$kgCO_2$-eq/kWh	Hill et al (2021) - DBEIS [158]
Climate Impact of Grid Electricity	119.75	$kgCO_2$-eq/tonne feed	-
Parasitic Load per tonne Feedstock	96.69	kWh/tonne feed	-
Climate Impact of Parasitic Load	20.05	$kgCO_2$-eq/tonne feed	-
Feed Transport Emissions	3.46	$kgCO_2$-eq/tonne feed	Hill et al (2021) – DBEIS [158]
Total Climate Impact (AD)	23.51	$kgCO_2$-eq/tonne feed	-

Table 24: GHG Inventory for Scenario 2: Gas-to-Grid

Data	Value	Unit	Source
Scenario 2 - Upgrading to Biomethane for Injection to Grid for Space Heating			
Upgrading Energy	0.25	kWh/m^3	Singh et al (2011) [111]
Upgrading Energy	49.44	kWh/tonne feed	Singh et al (2011) [111]
Climate Impact			
Conversion Factor (Biomethane)	0.00521	$kgCO_2$-eq/kg biomethane	Hill et al (2021) - DBEIS [158]
Conversion Factor (Natural Gas)	0.20297	$kgCO_2$-eq/kWh	Hill et al (2021) – DBEIS [158]
Climate Impact of Natural Gas	404.86	$kgCO_2$-eq/tonne feed	-
Parasitic Load per tonne Feedstock	137.31	kWh/tonne feed	-
Climate Impact of Parasitic Load	28.99	$kgCO_2$-eq/tonne feed	-
Feed Transport Emissions	3.46	$kgCO_2$-eq/tonne feed	Hill et al (2021) – DBEIS [158]
Total Climate Impact (AD)	32.45	$kgCO_2$-eq/tonne feed	-

Table 25: GHG Inventory for Scenario 3: HGV Fuel

Data	Value	Unit	Source
Scenario 3 - Upgrading to Biomethane and Compression for HGV Fuel			
Upgrading Energy	49.44	kWh/tonne feed	Singh et al (2011) [111]
Compression Energy	69.22	kWh/tonne feed	Singh et al (2011) [111]
Climate Impact			
Conversion Factor (Biomethane)	0.10625	$kgCO_2$-eq/MJ biomethane	Hill et al (2021) - DBEIS [158]
Conversion Factor (Diesel)	73.57	$kgCO_2$-eq/MJ diesel	Bates et al (2014) [154]
Climate Impact of Diesel	518.38	$kgCO_2$-eq/tonne feed	-
Parasitic Load per tonne Feedstock	202.32	kWh/tonne feed	-
Climate Impact of Parasitic Load	42.36	$kgCO_2$-eq/tonne feed	-
Feed Transport Emissions	3.46	$kgCO_2$-eq/tonne feed	Hill et al (2021) – DBEIS [158]
Total Climate Impact (AD)	42.36	$kgCO_2$-eq/tonne feed	-

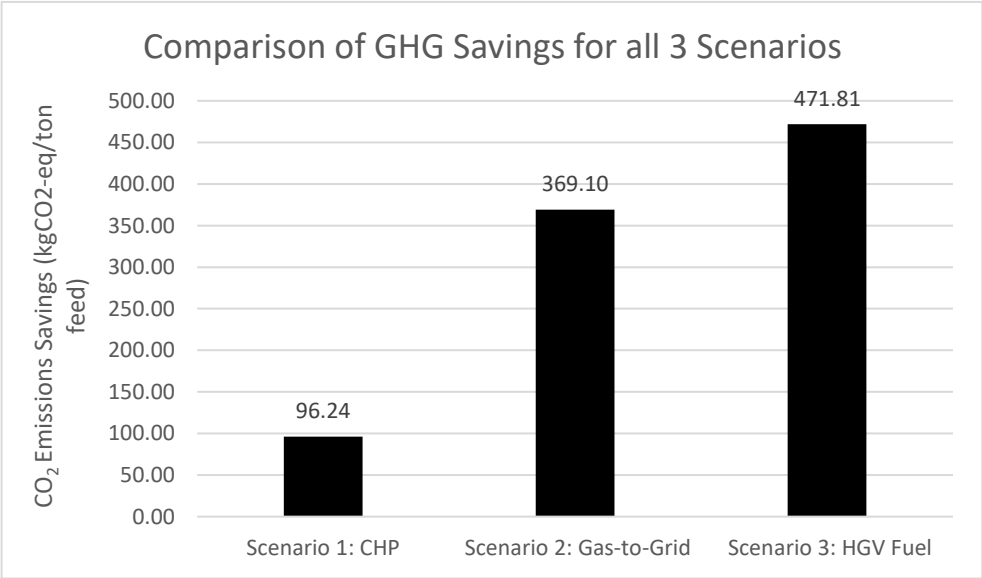


Figure 24: Comparison of GHG Emissions Savings for all 3 Utilisation Scenarios

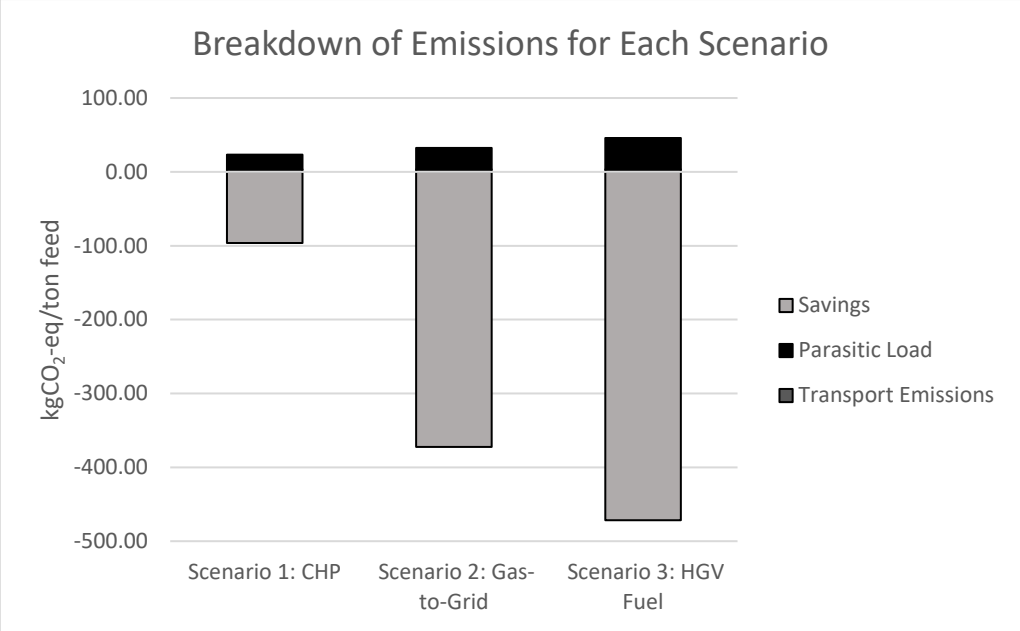


Figure 25: Breakdown of GHG Emissions for all 3 Utilisation Scenarios

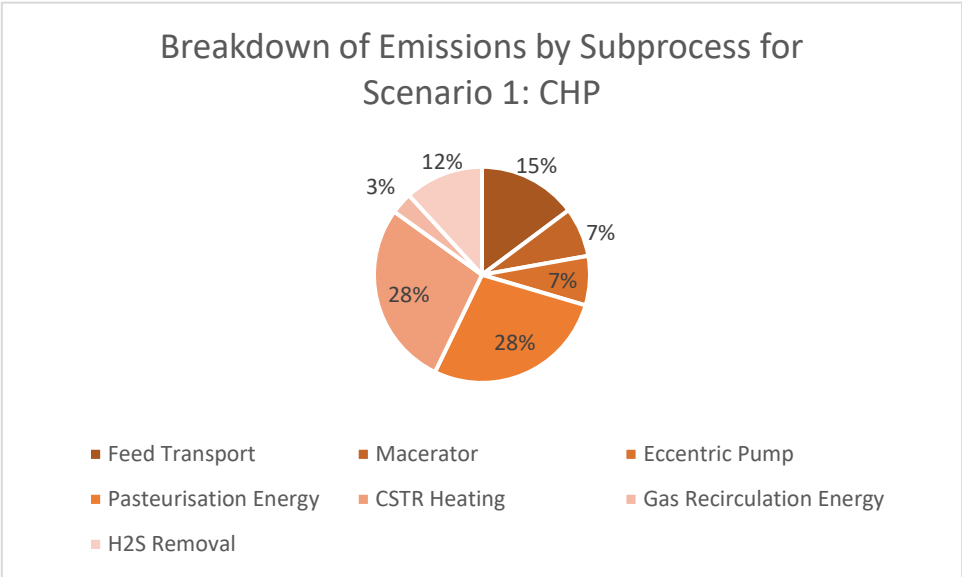


Figure 26: Breakdown of GHG Emissions for Each Subprocess for Scenario 1

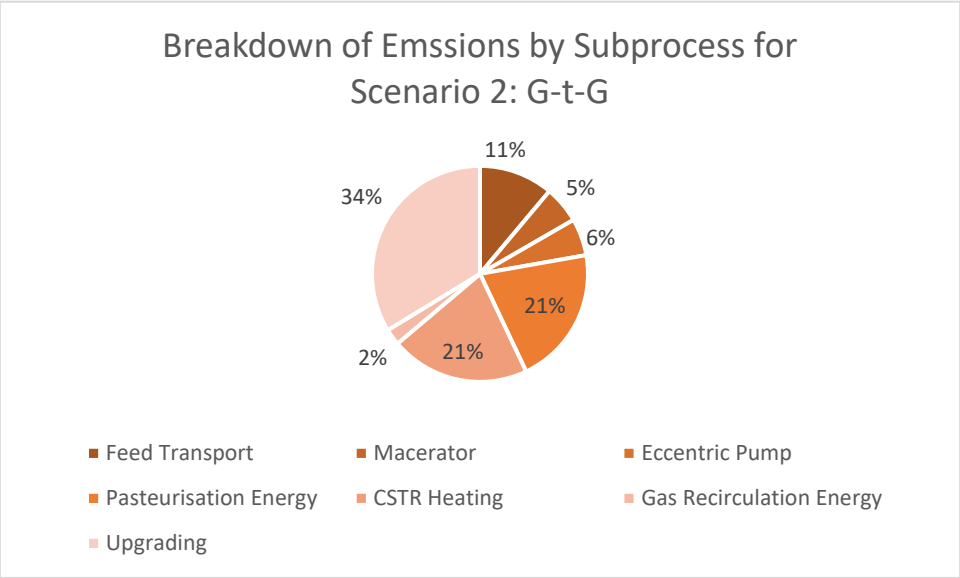


Figure 27: Breakdown of GHG Emissions for Each Subprocess for Scenario 2

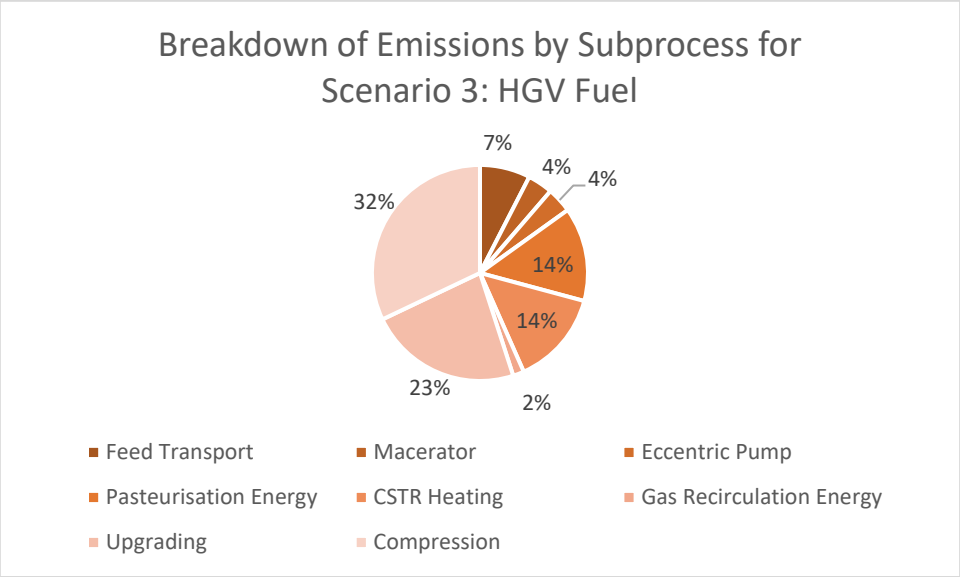


Figure 28: Breakdown of GHG Emissions for Each Subprocess for Scenario 3

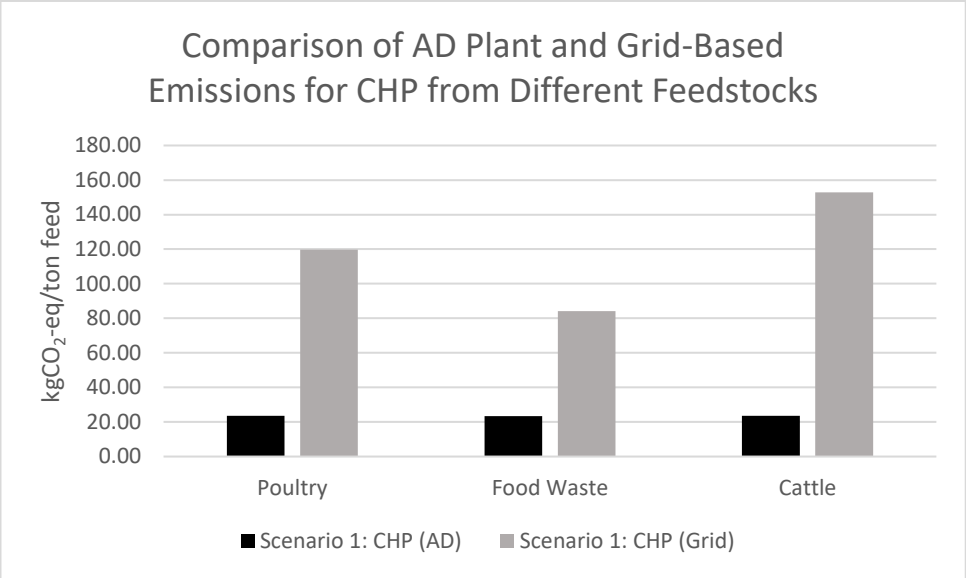
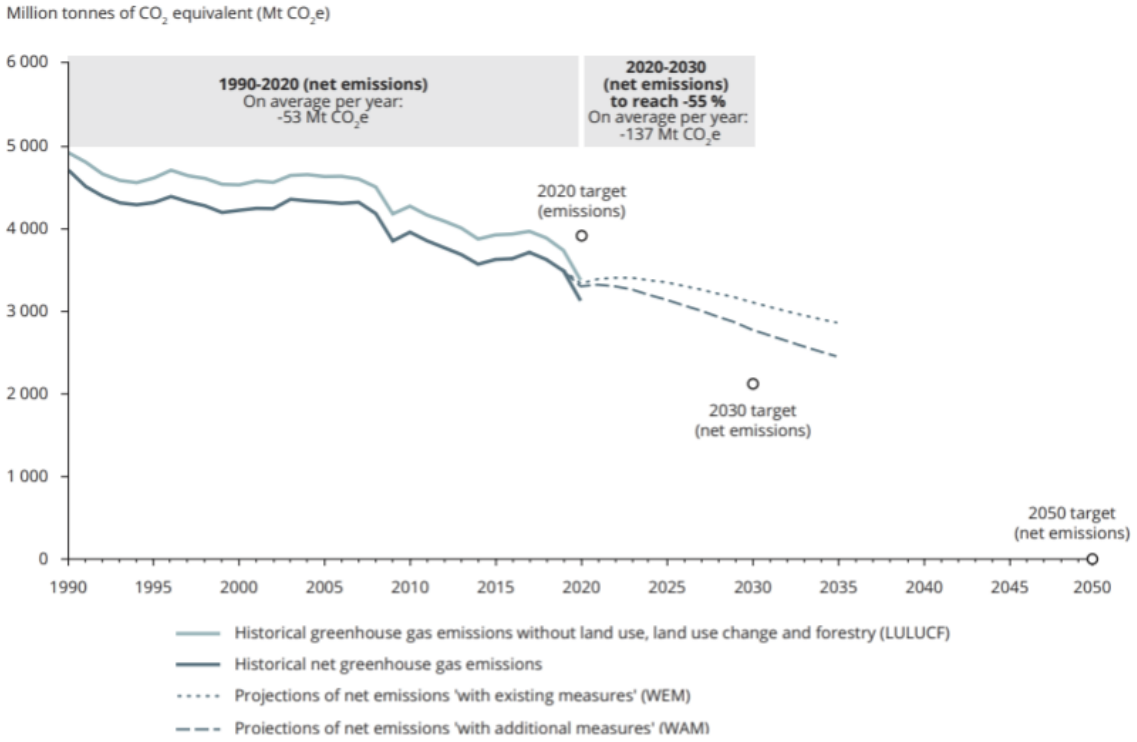


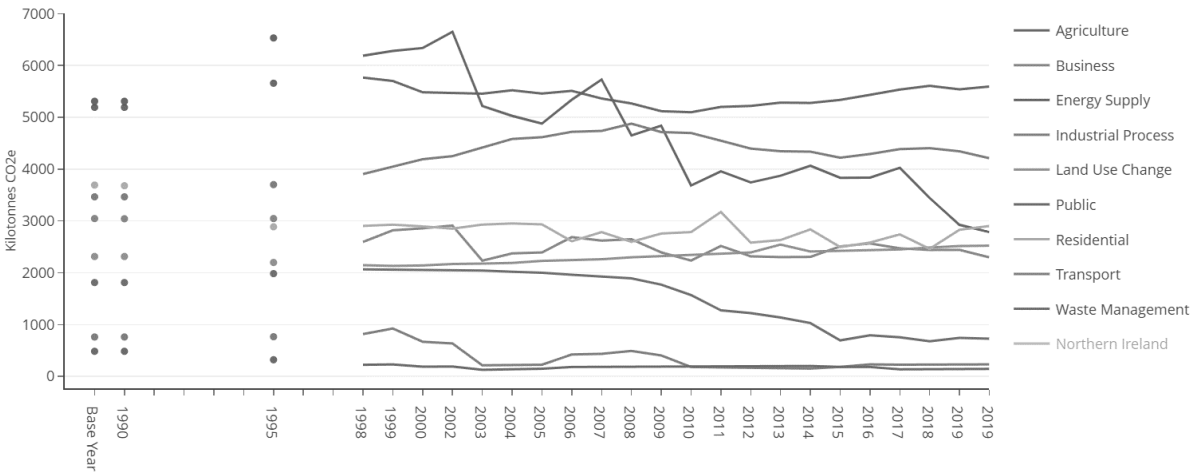
Figure 29: Effect of Feedstock on GHG Emissions

Appendixes

Appendix A: Historical trends and projections of greenhouse gas emissions [10]



Appendix B: Global GHG emissions by economic sector [13]



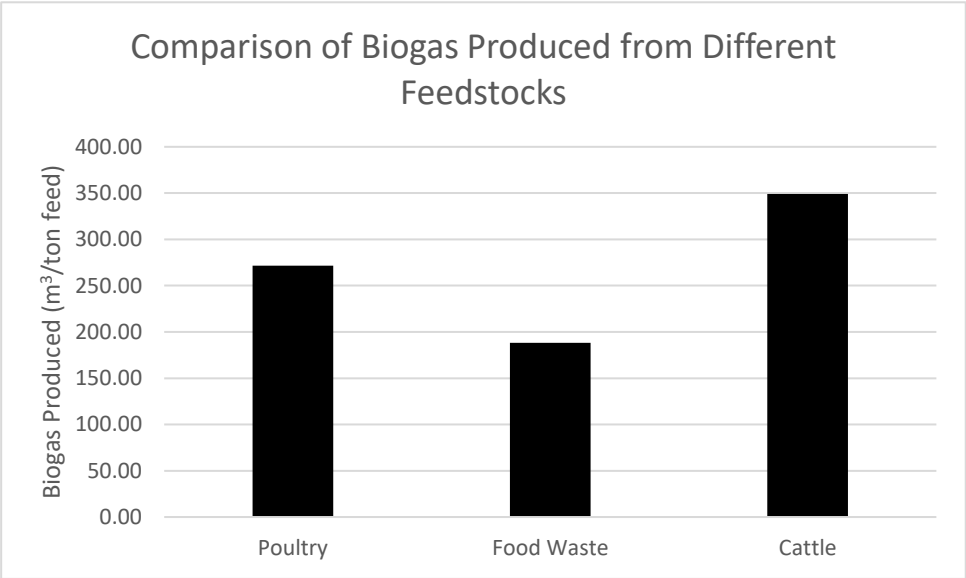


Figure 30: Effect of Feedstock on Biogas Production

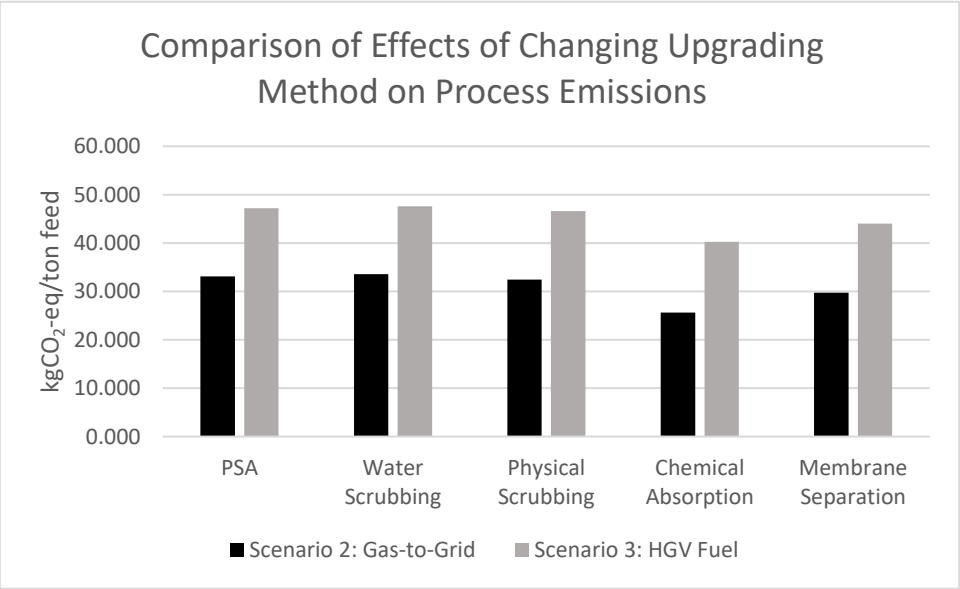


Figure 31: Effect of Upgrading Method on GHG Emissions

Appendix C: ROC/MWh Banding Levels from 1 July 2015 [32]

Technology	2014/15	2015/16	2016/17
Advanced gasification pyrolysis	2	1.9	1.8
Anaerobic digestion (≤ 500kW)	4	4	4
Anaerobic digestion (500kW – 5MW)	3	3	3
Anaerobic digestion (above 5MW)	2	1.9	1.8
Co-firing (low-range)	0.3	0.5	0.5
Co-firing (mid-range)	0.6	0.6	0.6
Co-firing (high-range)	0.9	0.9	0.9
Co-firing (low-range) with CHP*	0.8		Band closed to new accreditations from 1 October 2015
Co-firing (mid-range) with CHP*	1.1		Band closed to new accreditations from 1 October 2015
Co-firing (high-range) with CHP*	1.4		Band closed to new accreditations from 1 October 2015
Co-firing of regular bioliquids	0.3	0.5	0.5
Co-firing of regular bioliquids with CHP*	0.8		Band closed to new accreditations from 1 October 2015
Co-firing of energy crops (low-range)**	Band closed		
Co-firing of energy crops (low-range) with CHP**	Band closed		
Biomass conversion	1	1	1

Appendix D: NTS Specification (National Grid)

Content or Characteristic	Value
Hydrogen sulphide (H2S)content	≤5 mg/m3
Total sulphur content (including H2S)	≤50 mg/m3
Hydrogen content	≤0.1% (molar)
Oxygen content	≤0.2% (molar)
Impurities	shall not contain solid or liquid material which may interfere with the integrity or operation of pipes or any gas appliance (within the meaning of regulation 2(1) of the 1994 Regulations) which a consumer could reasonably be expected to operate
Hydrocarbon dewpoint and water dewpoint	shall be at such levels that they do not interfere with the integrity or operation of pipes or any gas appliance (within the meaning of regulation 2(1) of the 1994 Regulations) which a consumer could reasonably be expected to operate
Wobbe Number (WN)	(i) ≤51.41 MJ/m3, and (ii) ≥47.20 MJ/m3
Incomplete combustion factor (ICF)	≤0.48
Sooting index (SI)	≤0.60

Appendix E: EPA – DQPM [82]

Highest score ← → Lowest score

Indicator		1	2	3	4	5 (default)
Flow reliability		Verified[1] data based on measurements	Verified data based on a calculation **or** non-verified data based on measurements	Non-verified data based on a calculation	Documented estimate	Undocumented estimate
Flow Representativeness	**Temporal correlation**	Less than 3 years of difference[2]	Less than 6 years of difference	Less than 10 years of difference	Less than 15 years of difference	Age of data unknown or more than 15 years
	Geographical correlation	Data from same resolution **and** same area of study	Within one level of resolution **and** a related area of study[3]	Within two levels of resolution **and** a related area of study	Outside of two levels of resolution **but** a related area of study	From a different or unknown area of study
	Technological correlation	All technology categories[4] are equivalent	**Three** of the technology categories are equivalent	**Two** of the technology categories are equivalent	**One** of the technology categories is equivalent	**None** of the technology categories are equivalent
	Data collection methods	Representative data from >80% of the relevant market[5], over an adequate period[6]	Representative data from 60-79% of the relevant market , over an adequate period **or** representative data from >80% of the relevant market, over a shorter period of time	Representative data from 40-59% of the relevant market, over an adequate period **or** representative data from 60-79% of the relevant market, over a shorter period of time	Representative data from <40% of the relevant market, over an adequate period of time **or** representative data from 40-59% of the relevant market, over a shorter period of time	Unknown **or** data from a small number of sites **and** from shorter periods

Appendix F: Modified Traffic Light Matrix

Criteria	Description	Traffic light ranking	Colour
Technological representativeness	The data should be representative of current waste technologies or technologies likely to be used in Ireland. Examples include air emission data from the proposed waste incineration facility proposed at Poolbeg, Dublin.	Data from current Irish waste management facility	
		Data from technology likely to be used in Ireland	
		Data from technology with small chance of being used in Ireland	
Temporal representativeness	The age of the data will be recorded and assessed, with a preference for more recent data sources.	Data from last 5 years	
		Data from last 5 to 10 years	
		Data older than 10 years	
Geographical representativeness	The data will preferentially come from Irish data sources. When Irish data is not available, data will be sought from European sources and other international sources.	Irish data	
		UK and European data	
		Other international data	
Accuracy	The data should accurately represent the waste management process e.g. emission data is actual data from a facility.	Data is clearly representative of Irish waste management	
		Data does not represent Irish waste management, but appears to be similar process with some level of accuracy	
		Data accuracy is unclear	
Precision	A measure of the variability of the data points used to derive emissions, e.g. low variance = high precision, usually in reference to direct measurements of data. Direct measurements are not likely to be included in the model, but traffic lighting approach will still be used to compare the variability of the different data sources.	Low variance between minimum and maximum values from data sources	
		Medium variance between minimum and maximum value from data sources	
		High variance between minimum and maximum values	

Appendix G: Feedstock Traffic Lighting

Refence	Description	Representativeness	Representativeness	Geographical I	Accuracy	Precision	Score
[85]	An technical report spanning 15-years of operation in the AD of SHW in Sweden						21
[63]	Simulation of the AD of cattle SHW in Ireland						21
[75]	A quantification of AD feedstocks in NI						23
[87]	Lab-based analysis of the AD of poultry feedstock in Brasil						14
[86]	Lab-based study of the AD of SHW in Sweden						14
[63]	Lab-based study of the AD of poultry waste in Ireland						23

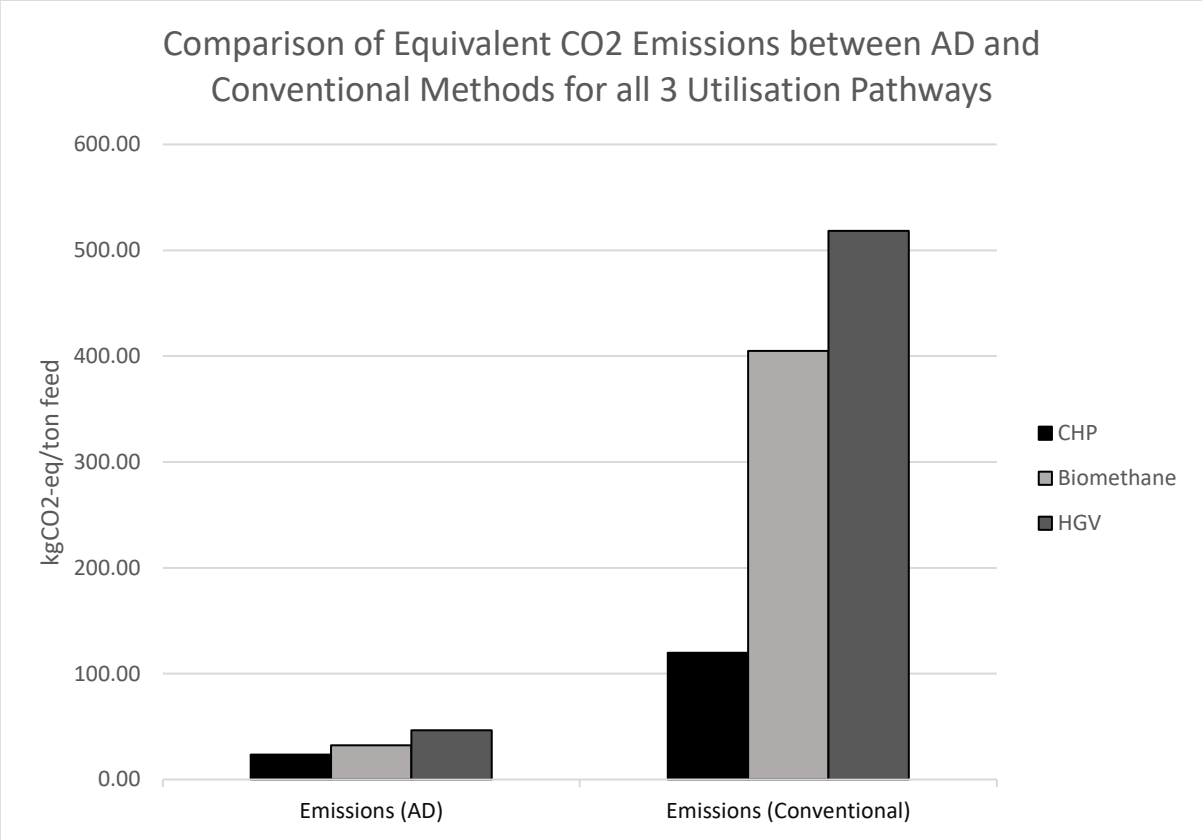
Appendix H: Characterisation of biogas from the AD of various FW [36]

Waste category	Waste amount: t/year	TS: %	VS: % of TS	Methane (CH_4) yield: m^3N CH_4/t VS	Methane yield average: m^3N CH_4/t VS	Methane content: %	Methane content average: %	Biogas: m^3/year	Methane: m^3/year
Household[a]	244 043	10	80	350–480	415	70–80	75	10 802 999	8 102 249
Sewage sludge (2–5% VS)[b]	38 566	2·5	65	140–210	175	60	60	182 789	109 673
Retail food[c]	35 700	36	66·5	200–440	320	60	60	4 558 176	2 734 905
Catering[d]	4140	29	95	467–529	498	60	60	946 673	568 003
Food processing[e]	26 000	15	80	276–738	507	60	60	2 636 400	1 581 840
Slaughterhouse[f]	178 230	10	77	400–610	505	60	60	11 550 789	6 930 473
Dairy[g]	13 200	31·5	89·5	455–708	581·5	60–80	70	3 091 428	2 163 999
Drinks and distillery[g]	12 000	5	91	335–385	360	60	60	327 600	196 560
Dairy cattle manure[e]	10 000 000	8·5	80	110–240	175	55–75	65	183 076 923	119 000 000
Pig manure[e]	500 000	5·5	80	175–400	287·5	70–80	75	8 433 333	6 325 000
Chicken manure[e]	300 000	20	80	210–480	345	60–80	70	23 657 142	16 560 000
Grass silage[h]	1 937 170	20	91	341–483	412	60	60	242 094 606	145 256 763
							Total	491 358 862	309 529 470
							Approximation	491 million	309 million 310 million

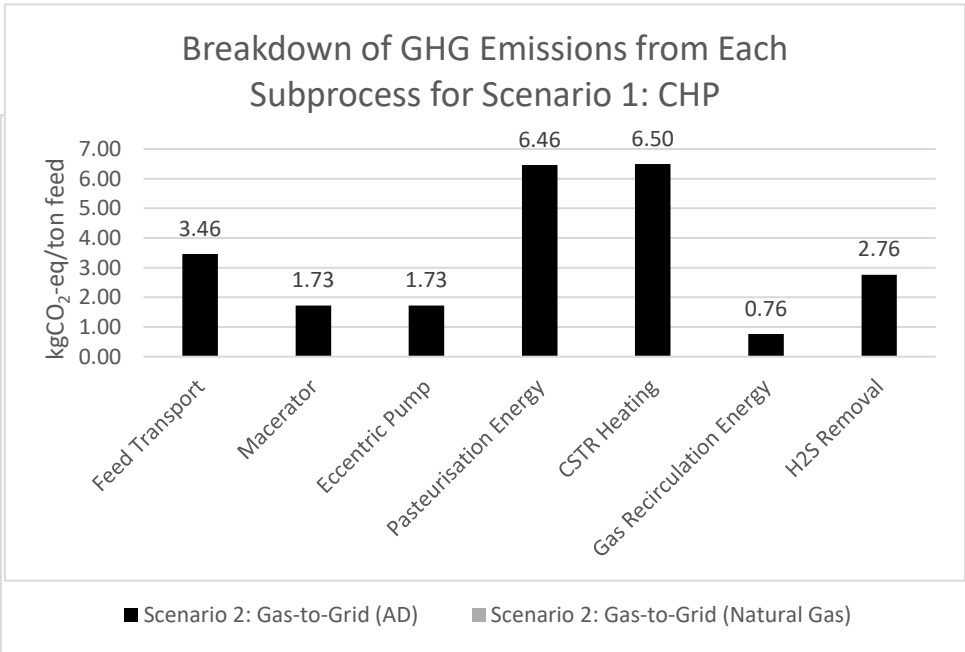
Appendix J: Results of the LCA from each scenario

Data	Value	Unit
Pathway 1 - Electricity to Grid from CHP Unit		
Emissions (AD)	23.51	$kgCO_2$-eq/tonne feed
Emissions (Conventional)	119.75	$kgCO_2$-eq/tonne feed
Savings	-96.24	$kgCO_2$-eq/tonne feed
Pathway 2 - Upgrading to Biomethane for Injection to Grid for Space Heating		
Emissions (AD)	35.76	$kgCO_2$-eq/tonne feed
Emissions (Conventional)	404.86	$kgCO_2$-eq/tonne feed
Savings	-369.10	$kgCO_2$-eq/tonne feed
Pathway 3 - Upgrading to Biomethane and Compression for HGV Fuel		
Emissions (AD)	50.77	$kgCO_2$-eq/tonne feed
Emissions (Conventional)	518.38	$kgCO_2$-eq/tonne feed
Savings	-467.61	$kgCO_2$-eq/tonne feed

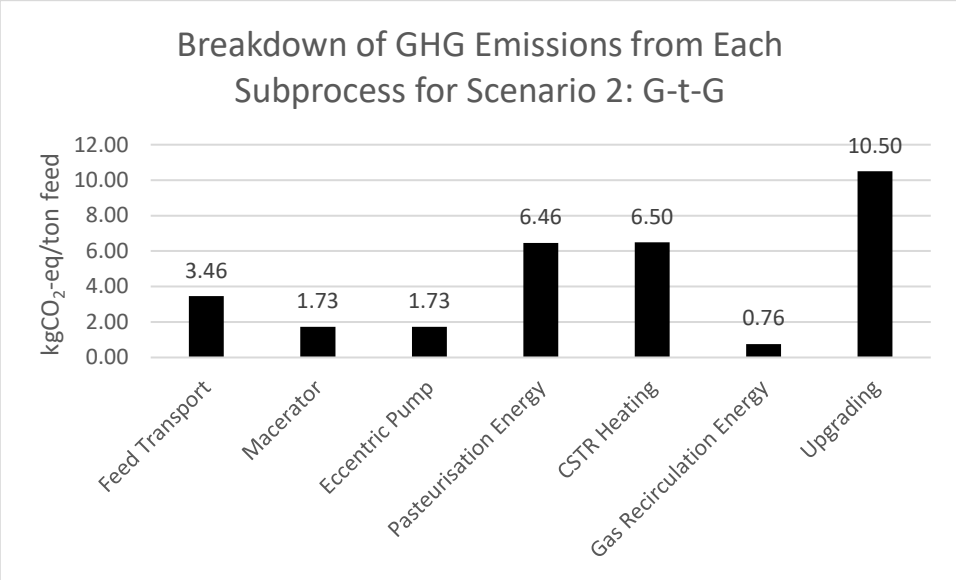
Appendix K:



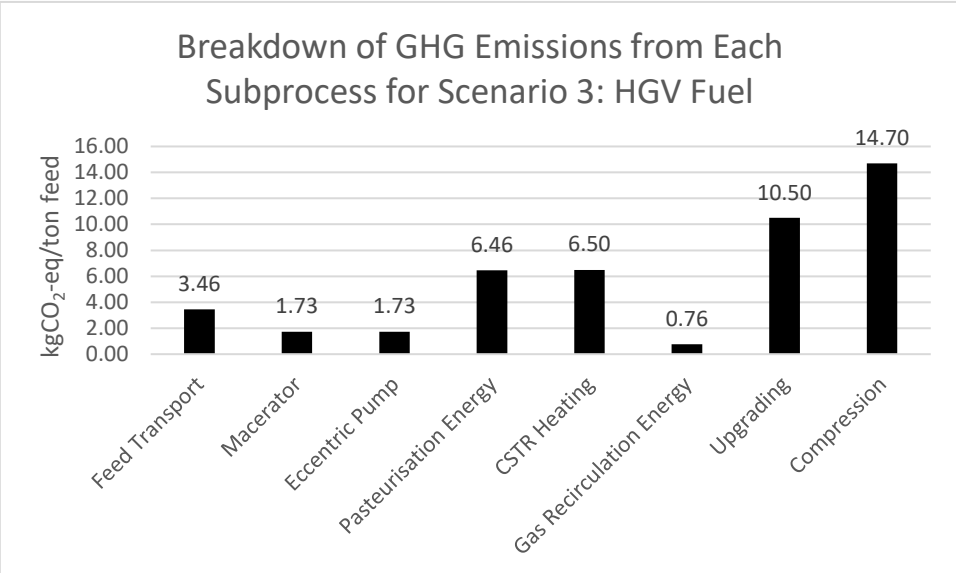
Appendix L



Appendix M:



Appendix N



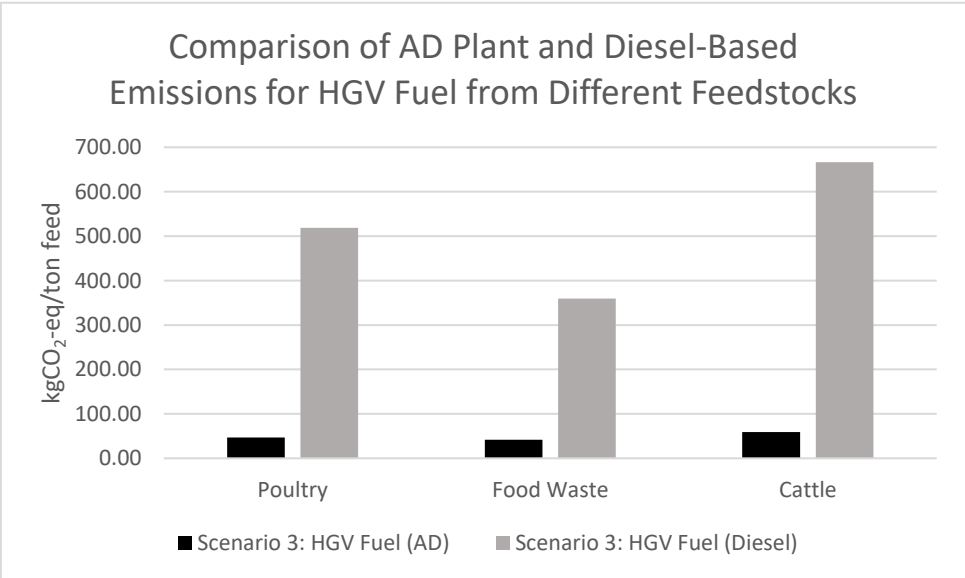
$$C_cH_hO_o + \left(c - \frac{h}{4} - \frac{o}{2}\right)H_2O \rightarrow \left(\frac{c}{2} + \frac{h}{8} - \frac{o}{4}\right)CH_4 + \left(\frac{c}{2} - \frac{h}{8} + \frac{o}{4}\right)CO_2$$

Figure 32: Buswell equation for the calculation of theoretical methane yiel

Appendix O

Data	Value	Unit
General AD Plant		
Feed Transport	3.46	$kgCO_2$-eq/tonne feed
Macerator	1.73	$kgCO_2$-eq/tonne feed
Eccentric Pump	1.73	$kgCO_2$-eq/tonne feed
Pasteurisation Energy	6.46	$kgCO_2$-eq/tonne feed
CSTR Heating	6.50	$kgCO_2$-eq/tonne feed
Gas Recirculation Energy	0.76	$kgCO_2$-eq/tonne feed
Scenario 1: CHP		
H_2S Removal	2.76	$kgCO_2$-eq/tonne feed
Scenario 2: G-t-G		
Upgrading	10.50	$kgCO_2$-eq/tonne feed
Scenario 3: HGV Fuel		
Upgrading	10.50	$kgCO_2$-eq/tonne feed
Compression	14.70	$kgCO_2$-eq/tonne feed

Appendix P



Appendix Q: Sensitivity analysis 2 – upgrading methods

	PSA	Water Scrubbing	Physical Scrubbing	Chemical Absorption	Membrane Separation	Unit
Energy Required	0.265	0.275	0.250	0.100	0.190	kWh/m^3
Pathway 2 - Upgrading to Biomethane for Gas-to-Grid						
G-t-G Emissions (AD)	33.136	33.591	32.452	25.619	29.719	$kgCO_2$-eq/tonne feed
G-t-G Emissions (Grid)	404.855	404.855	404.855	404.855	404.855	$kgCO_2$-eq/tonne feed
Savings	91.815	91.703	91.984	93.672	92.659	%
Pathway 3 - Upgrading to Biomethane and Compression for HGV Fuel						
HGV Fuel Emissions (AD)	47.201	47.621	46.571	40.272	44.052	$kgCO_2$-eq/tonne feed
HGV Fuel Emissions (Grid)	518.382	518.382	518.382	518.382	518.382	$kgCO_2$-eq/tonne feed
Savings	88.028	87.921	88.188	89.787	88.828	%

Sample Calculations

A heat and mass balance was performed in the determination of the CO_2 emissions of the AD plant over the process using Microsoft Excel. Using the AFBI plant as a reference, it was assumed that 20 tonnes per day (tpd) of feedstock would be processed. As there is no reaction taking place before the CSTR, a constant mass flow rate of 0.231kg/s was calculated through each of the pre-treatment phases before the reactor.

$$20tpd = \frac{20\ tons * 1000kg}{24hrs * 60mins * 60secs} = 0.231kg\ s^{-1}$$

As the feed requires pasteurisation at 70°C to eliminate pathogens, the energy required to raise the temperature of the feed was calculated by using

$$Q = \dot{m}c_p\Delta T$$

where Q = Power (kW), $\dot{m}$ = mass flow rate (kg/s), c_p = heat capacity (kJ/kg/K) and ΔT = temperature difference (K). Using the TS of the feed and assuming the remaining mass, the heating duty was calculated as follows:

$$Q = \dot{m} * \left(\frac{100 - TS}{100}\right) * c_{p\ water} * \Delta T$$

Assuming an ambient temperature of 15°C, TS of 44.9% and using the mass feed rate calculated above, this results in:

$$Q = 0.231 * \left(\frac{100 - 44.9}{100}\right) * 4.18 * (70 - 15) = 29.32kW$$

For the CSTR, it was assumed there is 100% conversion of volatile solids to biogas. From literature, the methane production rate (m^3/tonne VS) and %VS (dim.) of the feedstock are known, so combining them with the mass flow rate allows for calculation of the methane production rate:

$$\dot{V}_{methane} = 499.11\ m^3 ton\ VS^{-1} * \frac{20tpd}{24hr * 60min * 60sec} * 0.434VS\ ton^{-1}$$
$$= 0.0501m^3\ s^{-1}$$

The composition of biogas was assumed to be only CH_4 and CO_2 for simplicity of calculation, however there are trace amounts of other compounds such as H_2S which must be removed. To find the overall production rate of biogas, the amount of CO_2 was calculated as follows:

$$\dot{m}_{methane} = \dot{V}_{methane} * \rho_{methane} = 0.0501 * 0.657 = 0.032923 kg\ s^{-1}$$

Assuming a mixture of 65% methane and 32% carbon dioxide:

$$\dot{m}_{CO_2} = \frac{\dot{m}_{CH_4}}{65} * (100 - 65) = 0.034442 kg\ s^{-1}$$

$$\dot{V}_{CO_2} = \frac{\dot{m}_{CO_2}}{\rho_{CO_2}} = 0.017395 m^3\ s^{-1}$$

Total biogas production rate:

$$\dot{V}_{biogas} = \dot{V}_{CH_4} + \dot{V}_{CO_2} = 0.0695 m^3\ s^{-1}$$

$$\dot{m}_{biogas} = \dot{m}_{CH_4} + \dot{m}_{CO_2} = 0.06938 kg\ s^{-1}$$

It is assumed that there is no loss of mass inside the digester, therefore the digestate production rate is:

$$\dot{m}_{digestate} = 0.231 - 0.06938 = 0.162 kg\ s^{-1} = 14.004 tpd$$

As the amount of H_2S produced is negligible, no mass balance is carried out for the hydrogen sulfide removal column. Inside the CHP unit a simple combustion reaction takes place:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

It was assumed that there was no incomplete conversion and that all CH_4 is converted to CO_2. To calculate the O_2 required, a simple molar calculation was carried out:

$$\dot{n}_{CH_4} = \frac{\dot{m}_{CH_4}}{MW_{CH_4}} = \frac{231 g s^{-1}}{16.04 g mol^{-1}} = 2.052 mol\ s^{-1}$$

$$\dot{n}_{O_2} = 2\dot{n}_{CH_4} = 4.105 mol\ s^{-1}$$

$$\dot{m}_{O_2} = 2\dot{n}_{O_2} * MW_{O_2} = 0.131 kg\ s^{-1}$$

As the CO_2 from the digester is not involved in the reaction, it is assumed that the total CO_2 effluent from the CHP unit is the sum of the CO_2 produced in the digester and the CO_2 produced in the CHP unit. An electrical efficiency of 33% is used in the calculation of the energy output of the CHP, with a 43% thermal efficiency and 24% losses. The electricity produced by the CHP unit is calculated as follows:

$$P = \dot{V}_{biogas} * CV_{biogas} * 1000 * 0.33 = 0.0695 * 22 * 1000 * 0.33 = 504.80 kW$$

(CV_{biogas} is the calorific value of biogas in MJ m^{-3}, therefore the result is multiplied by 1000 to convert to kW).

The thermal energy gains and losses are calculated in the same way, using their respective efficiency factors.

To compare the emissions of this process compared to grid emissions, conversion factors were used to calculate the kg_{CO2}-eq for the electricity produced. Assuming 330 days of operation a year:

$$E = \frac{P * t}{\dot{m}_{feed}} = \frac{504.80 * 24 * 330}{7300tpa} = 547.67kWh\ ton^{-1} feedstock$$

Parasitic load was calculated by summing the total electricity and heat inputs to the process:

$$Parasitic\ Load_{electricity} = 30.3kW = 32.9kWh\ ton^{-1}\ feedstock$$

$$Parasitic\ Load_{heat} = 58.8kW = 63.8kWh\ ton^{-1}\ feedstock$$

Parasitic load emissions were calculated using the grid equivalent conversion factors:

$$PL\ Emissions_{electricity} = 32.9 * 0.21233 = 12.95kg_{CO_2eq} ton^{-1} feedstock$$

$$PL\ Emissions_{heat} = 63.8 * 0.20297 = 10.55kg_{CO_2eq} ton^{-1} feedstock$$

$$PL\ Emissions_{total} = 12.95 + 10.55 = 23.49kg_{CO_2eq} ton^{-1} feedstock$$

$$\begin{aligned} Emissions_{biogas} &= E * Conversion\ Factor + PL\ Emissions_{total} \\ &= 547.67kWh\ ton^{-1} feedstock * 0.00022kg_{CO_2eq}\ kWh^{-1} + 23.51 \\ &= 23.51kg_{CO_2eq} ton^{-1} feedstock \end{aligned}$$

Comparatively

$$\begin{aligned} Emissions_{grid} &= E * Conversion\ Factor \\ &= 547.67kWh\ ton^{-1} feedstock * 0.21233kg_{CO_2eq}\ kWh^{-1} \\ &= 119.75kg_{CO_2eq} ton^{-1} feedstock \end{aligned}$$

This method was repeated to calculate emissions savings for G-t-G and HGV Fuels.